

In Search of Working Time?

Hours Constraints, Firms and Mobility*

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Abstract

Can workers find a job with their ideal working hours? This paper provides novel empirical evidence on hours constraints, i.e. barriers for workers to work their desired hours at a given wage rate, by linking self-reported hour preferences from large-scale survey data with administrative employer-employee data between 2003 and 2023. Twenty percent of French salaried workers report wanting to increase their hours at their given wage. Leveraging the panel dimension of my data, I show that constrained workers experience larger increases in hours and earnings than other workers by switching employers. However, most constrained workers remain unable to adjust their hours to their desired level within 3 years after their report. Next, I develop a revealed preference method to quantify welfare effects associated with constraints and find that workers would on average accept a 10% reduction in hourly wages to work in a job offering their desired number of hours. The results highlight the important role of hours worked as a job amenity in shaping labor market sorting.

Keywords: Working Hours, Labor Supply, Hours Constraints, Part-Time Work.

JEL Codes: J22, J31, C81

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1 Introduction

Both a source of income and disutility, working hours are a central element of workers' lives (Dube et al., 2022; Mas, 2025). However, most workers cannot freely choose their hours since employers have a clear interest in the hours worked by their employees, as first emphasized by Lewis (1967). This implies that several workers may be *constrained* in jobs where they cannot work their preferred number of hours. When unable to choose their hours, a crucial question is whether they can, and at what cost, circumvent hours constraints to find a job that better suits their preferences.

This paper provides novel empirical evidence on the topic of hours constraints, focusing on the dynamics of the phenomenon. A large literature has documented the prevalence of hours constraints, but little is known about their persistence and welfare impacts. I ask two related questions: can most constrained workers reach their desired hours through labor market mobility, and how much of their hourly wage are they willing to give up to do so? Combining self-reported hours preferences from the French Labour Force Survey with panel employer-employee data, I can track workers after their report and observe their changes in hours, and earnings, over the following years. Mobilizing employer-to-employer moves in a revealed preference framework, I quantify the welfare impacts of hours constraints on workers. The richness of my data finally allows me to document other aspects of hours constraints, such as the sorting of constrained workers across firms and occupations.

Hours constraints have received significant attention in the labor literature (see Kahn and Lang (2001) for an introduction). Empirical studies consistently find that a large proportion of workers are not working their desired hours, although the direction of constraints varies across countries. Jarosch et al. (2025) show in a recent paper that German and British workers tend to be *overworked*, or constrained from below, while US workers are mostly *underworked* or constrained from above. In the German case, more than two thirds of workers report wanting to work fewer hours. France is closer to the US case : evidence from the *Enquête Emploi*, the French Labour Force Survey, indicates that 20% of salaried workers would work more hours at their given wage rate, the most common type of constraints in France. The magnitude of the shares across countries points to hours constraints as a widespread labor market phenomenon.

A first question concerns the origins of constraints: what prevents workers from working their preferred hours? Working hours are heavily regulated, particularly in France¹, which implies that employers may have to bear additional (overtime) costs to increase hours of their employees. This restriction yet does not apply to the frontier between part-time and full-time work, which is at the

¹See Breda et al. (2025) for more on the French working time regulation.

heart of this paper. At least for involuntary part-time workers, working time regulation cannot be the sole reason why they cannot work their preferred hours. Early research on the topic ([Altonji and Paxson, 1986](#); [Dickens and Lundberg, 1993](#)) rather emphasized the importance of occupation in hours practices. Certain jobs, e.g. train drivers or school teachers, are associated with specific hours requirements that may leave little room for adjustment. However, recent work has rather focused on the role of firm-specific features, following the idea that hours worked likely depend on each firm's organization and technology. [Kahn and Lang \(2001\)](#) provide a broad discussion of the potential sources of firms' preferences over hours, e.g. highlighting the role played by coordination in team production contexts, as in [Rosen \(1986\)](#). [Labanca and Pozzoli \(2023\)](#) build on this idea to rationalize hours constraints as compensated by wage differentials arising from productivity gains. Another line of argument connects hours constraints to monopsony power ([Manning, 2003](#)), in that firms may be able and have an interest in keeping some of their workers underemployed to freely adjust their hours when the activity fluctuates ([Bell and Blanchflower, 2021](#)). [Lachowska et al. \(2026\)](#) includes the most comprehensive theoretical framework to date for the study of constraints, using an extension of the Lewis-Rosen model ([Lewis, 1969](#); [Rosen, 1974](#)) under imperfect competition. Their findings point to the existence of a job ladder on wages and hours, i.e. a hierarchical ranking of employers based on job desirability, as the source of hours constraints, and conclude that long-hour jobs are too costly for most employers.

A second question concerns the welfare implications of hours constraints. Survey data suggest the distance to desired hours may be large, particularly for part-time workers who mostly want to increase their hours to a full-time schedule. This implies that constrained workers may be far from their labor supply curve and would accept a large wage reduction to close the gap. [Lachowska et al. \(2026\)](#) take a revealed preference approach to estimate this gap and find that workers would on average require a 12% higher wage to be as well off as they would be while working their ideal hours. They also find a 15% average distance between actual and desired hours, consistent with survey data. Their methodology reflects the magnitude of welfare effects associated with constraints at the labor market level, but it does not rely on actual moves of workers who are in a constrained situation, which is precisely what my data allow.

The study of hours constraints has been hampered by a lack of suitable data. Preferences over hours are hard to identify, and aside from a few exceptions, e.g. [Labanca and Pozzoli \(2023\)](#) use a firm-level proxy based on the within-firm dispersion of hours, the most common approach relies on surveys. In addition, surveys that directly elicit hours preferences are rarely panel data or linked to firm-level information, with the notable exception of the data used in [Jarosch et al. \(2025\)](#), and their measures of hours and earnings are likely imperfect, relying on self-reports rather than administrative records.

In this paper, I construct a novel dataset by linking self-reported hour preferences from a large-scale survey with panel employer-employee data. I exploit the presence of establishments' identifiers in the French Labour Force Survey, conducted by Insee², and use them to link survey data to administrative records. As there are no common individual identifiers between both sources, I adopt a "1-to-1" matching approach based on variables such as age, gender or birth department combined with the firm identifier. This method allows me to retain information for a majority of surveyed salaried workers (67%, 435,472 individuals) between 2003 and 2023, capturing their self-reported hour preferences over six quarters and their administrative job characteristics, including paid hours and labor earnings, over a large number of years. By exploiting the panel structure of the administrative data, I can study the labor market trajectories of constrained workers and examine the welfare effects associated with their job changes. This unique linkage extends beyond the scope of hours constraints and contributes to bridging survey and administrative sources for economic research. The remainder of this paper is organized in three steps.

The first step examines the employment context of workers who report hours constraints to improve our understanding of the phenomenon. Constrained workers are concentrated at the bottom of the (hourly) wage distribution and primarily seek to increase earnings through additional work. This is especially the case for involuntary part-time workers, who report being unable to switch to a full-time schedule and are constrained in particular low-wage jobs. On the other hand, full-time constrained workers work a similar number of hours than unconstrained workers and seem to mostly differ in terms of preferences rather than actual working conditions. Occupational sorting emerges as a major factor for involuntary part-time work, as concerned workers are heavily concentrated in few low-wage occupations notoriously intensive in part-time jobs. Firm sorting also plays an important role, as previously hinted in the literature ([Kahn and Lang, 2001](#); [Lachowska et al., 2026](#)) but rarely shown empirically. Using an AKM ([Abowd et al., 1999](#)) decomposition to identify firm effects in both hours and wages, I find that constrained workers are overrepresented in firms offering low hours and low wages, consistent with the existence of a job ladder. Yet, there remains non-negligible variation in hours within firms, e.g. most involuntary part-time workers have at least one full-time worker employed in the same workplace and occupation as them. Tackling this within-firm heterogeneity, I provide suggestive evidence of worker segmentation over access to full-time jobs associated with gender and experience.

The second and main part of the paper provides the first dynamic study of hours constraints at the worker-level. Leveraging the panel dimension of my data, I track the labor market trajectories of constrained workers in the years following their survey report. Conditional on characteristics, constrained workers are not more likely than their unconstrained counterparts to switch employ-

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ers, yet when they move, they benefit more from mobility in terms of hours and earnings. Using an event-study design, I estimate that constrained part-time workers increase their hours and earnings through mobility by roughly 6% more than unconstrained movers, and by 8% and 4% more than non-movers, respectively. This effect is concentrated among part-time workers, as I find no evidence of differential mobility outcomes in the full-time group. The richness of my data also allows me to provide a thorough description of adjustment mechanisms. Workers are more likely to adjust their hours by switching occupation or moving to firms with long-hour policies, consistent with the prior identified role of occupational and firm rigidities. However, mobility does not appear sufficient to close the hours gap, as most constrained workers are unable to reach their self-reported preferred hours in the short-run after their report.

The third step develops a method to quantify workers' willingness to pay (WTP) in hourly wages to work their preferred hours, providing a welfare measure of hours constraints. Since the primary motivation for wanting additional hours is higher monthly earnings, the WTP is best interpreted as the hourly wage reduction a worker is willing to accept in exchange for an increase in monthly earnings. The method, adapted from [Le Barbanchon et al. \(2020\)](#) on the relation between wage and commute, takes advantage of workers' self-reported preferred hours and of their employer-to-employer moves. Workers start from a job with a given wage and hours and report their desired hours at that wage (as in the Labour Force Survey). A simple revealed preference implies that workers would only switch employers if the move raises their utility. I then estimate the main utility parameter by recovering the iso-utility curve that best rationalizes employer-to-employer moves given this criterion. I solve the optimization problem using a grid search algorithm and obtain an individual WTP estimate for each constrained worker who switch firms after they report. The average WTP is equal to 10% of workers' current wage, consistent with estimates by [Lachowska et al. \(2026\)](#) in the US. This answers the second question asked above: workers are willing to significantly reduce their hourly wages to increase their hours and earnings.

Related Literature. This work falls within a prominent, although specialized, literature on hour constraints. The theoretical foundations of hour constraints have been established in seminal work by [Rosen \(1974\)](#), then developed and refined ([Abowd and Card, 1987](#); [Dickens and Lundberg, 1993](#); [Kahn and Lang, 2001](#); [Bloemen, 2008](#)). This literature intends to provide a framework in order to understand how hour constraints emerge and affect labor markets. Recent contributions to this literature relate hours constraints to the effects of imperfect competition on hours worked ([Manning, 2003](#); [Germain, 2025](#)). Specifically on the question of part-time work, [Devicienti et al. \(2020\)](#) discusses the monetary (dis)incentives for firms to offer part-time contracts. More closely related, [Carry \(2024\)](#) studies the impact of a minimum workweek reform in the case of France and find evidence of imperfect substitutability between part-time and full-time jobs. The main refer-

ence for this paper is [Lachowska et al. \(2026\)](#), who build on a revealed preference estimation to quantify welfare effects of constraints. I contribute to this literature by combining a micro-level measure of hours constraints with administrative panel data to reassess important findings of this literature, including welfare effects, in a more direct way.

This research also relates to a literature considering hours worked as a job amenity that influences workers' job choices and satisfaction. [Hwang et al. \(1998\)](#), [Lang and Majumdar \(2004\)](#), [Lavetti and Schmutte \(2016\)](#), [Sorkin \(2018\)](#), [Lamadon et al. \(2022\)](#), and [Dube et al. \(2022\)](#) have examined how hours worked affect workers' utility and job preferences. This paper adds to this body of work by estimating how much workers value their working hours.

This paper also relates to a macro literature documenting the countercyclical relation of underemployment to business cycles ([Borowczyk-Martins and Lalé, 2020](#); [Bell and Blanchflower, 2021](#); [Tuda, 2020](#)). This literature highlights underemployment as a transitory phenomenon, driven by employment adjustment to economic activity. With this paper, I contribute to this discussion by assessing the persistence of underemployment over a certain number of years. The result that most constrained workers remain unable to adjust their hours as wanted points to underemployment as a structural feature of the labor market.

Because of its relevance for the public economics literature, the relationship between hour constraints and labor supply responses to shocks has been largely explored. [Kahn and Lang \(1991\)](#) discusses the bias in labor supply elasticities due to the presence of hours constraints. [Chetty et al. \(2011\)](#) assess the implications of hours constraints for quasi-experimental estimation of labor supply elasticities. [Labanca and Pozzoli \(2022\)](#) show the unresponsiveness to tax changes in the context of hours constraints. This paper contributes to this literature by assessing the extent of hours rigidities in the labor market and whether they can be circumvented.

Lastly, this paper makes an important contribution to the measurement of true labor supply, to be distinguished from observed hours worked ([Pencavel, 2015](#)). The study of hours constraints complements extensive macroeconomic research ([Costa, 2000](#); [Bick et al., 2018, 2022](#)) that directly interprets trends in hours worked as evidence of labor supply changes. Recent research by [Jarosch et al. \(2025\)](#) also belongs to this literature by estimating the implied macroeconomic effects of relaxing constraints. In the same vein, another strand of literature (see [Bell and Blanchflower \(2021\)](#)) has focused on the empirical assessment of underemployment, focusing on different countries and discussing the potential origins of this phenomenon. This paper reinforces the claim that hours constraints are a major feature of labor markets and should be invoked in all discussions related to working time.

The rest of the paper proceeds as follows. Section 2 introduces a theoretical framework. Section 3 presents the institutional context and data, including the procedure to link the French Labour Force Survey to administrative data. Section 4 provides descriptive statistics on constrained work-

ers and studies the role of occupational and firm effects in shaping constraints. Section 5 tracks the evolution of constrained workers in the labor market, focusing on their ability to increase their hours and earnings. Section 6 estimates the workers' willingness-to-pay to reach their preferred hours in a revealed preference framework.

2 Model of Hours Constraints and Job Mobility

This section provides a theoretical framework for the study of hours constraints. Most of the framework is directly derived from the Lewis-Rosen model under imperfect competition introduced by [Lachowska et al. \(2026\)](#). The main difference is the introduction of heterogeneity in worker productivity θ_i , which intervenes in the employer's technology. This addition implies that an employer may have heterogeneous preferences on working hours for different workers, even if they have similar preferences. In the second part of the section, I introduce a dynamic extension of the Lewis-Rosen model and discuss conditions under which constrained workers are able to adjust their hours. This framework serves as an illustration to the empirical work realized in the further sections.

2.1 Static Bargaining

Workers are indexed by $i = 1, \dots, N$ and have preferences on earnings and hours represented by the utility function $u_i(\cdot)$, increasing in earnings and decreasing in hours. The (hourly) wage w is defined as the ratio between earnings and hours. For each worker, the optimal number of hours worked h^* is defined by the equality between the marginal rate of substitution (MRS) between earnings and hours and the observed wage:

$$\text{MRS}_{e,h}(e, h^*) = -\frac{\partial u_i(e, h)}{\partial h} / \frac{\partial u_i(e, h)}{\partial e} \Big|_{h=h^*} = w$$

This equality is a direct consequence of the utility maximization. If the MRS is superior to the wage, the cost of working one extra hour is superior to the gain, hence the worker is overemployed. Likewise, the worker is underemployed if the MRS is inferior to the wage.

Employers are indexed by $j = 1, \dots, J$ and have heterogeneous technologies inducing employer-specific revenue functions $R_j(\cdot)$. The revenue function is assumed increasing concave in hours worked, and depends on a worker-specific parameter θ_i introducing heterogeneity in worker productivity. As in [Carry \(2024\)](#), firms have heterogeneous technologies of production T_j and a maximum number of productive hours requirement z_j . The parameter T_j captures any specificity of the firm's production process and can be thought of as a TFP parameter. The profit function of employer j for a given match with worker i is $\Pi_j(h, \theta_i, w) = R_j(h, \theta_i) - wh$.

Following [Lachowska et al. \(2026\)](#), imperfect competition is incorporated through a firm-specific surplus k_j , positive for all j 's, that firms are able to extract. All firm-specific parameters are jointly drawn from a distribution $\Gamma(T, z, k)$, with T_j and k_j assumed independent. The bargaining problem³ that generates hours and wages at the equilibrium is given by:

$$(w_{ij}^b, h_{ij}^b) = \operatorname{argmax}_{w, h} u_i(e, h) \text{ s.t. } \Pi_j(h, \theta_i, w) = k_j \quad (1)$$

In the case of an interior solution ($h \leq z_j$), the number of hours that solves the optimization problem, h_{ij}^b , is defined by the equality between the MRS and the marginal revenue of hours⁴:

$$\text{MRS}_{e, h}(e, h) = \frac{\partial R_j(h, \theta_i)}{\partial h} \quad (2)$$

The number of hours worked in a given firm is generated by the properties of its revenue function R_j and the worker's productivity θ_i . Since the revenue function depends on this productivity, the number of hours worked in a given firm can vary between workers with identical preferences. If the marginal revenue of an additional hour of work is increasing in θ_i , bargained hours are increasing in θ_i , i.e. employers offer long-hour jobs to workers that they perceive as more productive.

The bargained wage w_{ij}^b is then derived as the average revenue per hour marked down by a profit margin that depends on the firm-specific surplus k_j ⁵:

$$w_{ij}^b = \frac{R_j(h_{ij}^b, \theta_i)}{h_{ij}^b} \left[1 - \frac{k_j}{R_j(h_{ij}^b, \theta_i)} \right] \quad (3)$$

Then, the bargained wage varies based on the firm-specific revenue function R_j , the firm-specific surplus k_j , and the worker's productivity θ_i .

Workers are constrained from above at the equilibrium if their MRS is lower than the bargained wage, as they would accept to work one more hour for less than the bargained wage. Using Equations 2 and 3, this leads to the condition:

$$\left. \frac{\partial R_j(h, \theta_i)}{\partial h} \right|_{h=h_{ij}^b} < \frac{R_j(h_{ij}^b, \theta_i)}{h_{ij}^b} \left[1 - \frac{k_j}{R_j(h_{ij}^b, \theta_i)} \right]$$

Hence, a given worker i in firm j is constrained from above if the marginal benefit of one ad-

³As discussed in [Lachowska et al. \(2026\)](#), this problem yields the same hours allocation that would be obtained in a Nash bargaining problem where the joint surplus of the match is maximized.

⁴See Appendix C.1 for the proof.

⁵See Appendix C.1 for the proof.

ditional hour, the gain in revenue for the firm, is lower than its marginal cost, the wage. While increasing hours is beneficial to the worker, the firm would lose from hiring the worker for one more hour, as the marginal product of this additional hour is too low as compared to the wage. This idea, already present in [Kahn and Lang \(2001\)](#), is at the heart of hours constraints: workers want to work more hours *at their current wage*, while their productivity is lower and lower due to the concavity of the revenue function. Following [Lachowska et al. \(2026\)](#), I write as α_{ij} the elasticity of the revenue function of firm j with respect to hours worked of worker i and assume it is constant⁶. With the profit margin $p_{ij} \equiv k_j / R_j(h_{ij}^b, \theta_i)$, hours constraints from above are observed if and only if :

$$\alpha_{ij} < 1 - p_{ij} \quad (4)$$

Equation 4 highlights that hours constraints are driven by demand-side characteristics: the revenue elasticity α_{ij} , which reflects the firm's efficiency in using one extra hour of work, and the profit margin p_{ij} . Since bargained hours and wages both vary with worker productivity θ_i , workers matched to the same firm but differing in productivity have different hours and wage outcomes, which affects their probability to be constrained from above.

2.2 Dynamic Lewis-Rosen Model

This section focuses on the ability for constrained workers to secure a better wage-hour bundle over time. To do this, I develop a dynamic extension of the Lewis-Rosen model, drawing on the sequential auction framework of [Postel-Vinay and Robin \(2002\)](#) and its adaptation in Appendix D.2 of [Lachowska et al. \(2026\)](#). In [Postel-Vinay and Robin \(2002\)](#)'s framework, firms make take-it-or-leave-it wage offers to workers and can counter outside offers received by their employees. Then, a worker moves to a poaching firm if the latter can offer a higher lifetime utility than the maximum the incumbent is willing to provide. I extend their model by making the job arrival rate endogenous to search intensity and varying along worker productivity to characterize under which conditions can workers adjust their hours.

Maximum Utility at Incumbent Firm. The bargained utility of worker i at firm j is denoted $u_{ij}^b = u_i(w_{ij}^b, h_{ij}^b)$, and is derived from the procedure presented in the previous section. Define $u_{ij}^{\max}$ as the maximum flow utility firm j can offer to worker i , i.e. the highest flow utility it can provide with a non-negative profit. This implicitly corresponds to a reduction of k_j , the firm-specific surplus associated with the job, to 0. As shown in Equation 2, bargained hours remain at the same level, $h_{ij}^{\max} = h_{ij}^b$, but the bargained wage increases after the markdown drops to zero in Equation 3, which implies that $u_{ij}^{\max} \geq u_{ij}^b$.

⁶This assumption is verified upon using an isoelastic revenue function, e.g. $R_j(h, \theta_i) = \theta_i T_j h^{\alpha_{ij}}$.

Value Function. An employed worker i at firm j chooses search intensity to find another job $s_i \geq 0$ at a flow cost $c(s_i)$. Job offers arrive to employed workers at a Poisson rate, with $\lambda > 0$:

$$\lambda(s_i, \theta_i) = \lambda \cdot \phi(\theta_i) \cdot s_i \quad (5)$$

where $\phi(\theta_i)$ is increasing in θ_i , capturing that high-productivity workers are more visible to or more easily matched with new firms. Each offer is a draw of a new firm j' from the distribution Γ .

Let V_{ij} denote the lifetime utility of a worker i being employed by firm j , and V_i^u the value of unemployment. The poaching utility $u_{i\ell,j}^{poaching}$ that firm ℓ has to offer to hire worker i from firm j solves: $V_{ij}(u_{i\ell,j}^{poaching}) = V_{ij}(u_{ij}^{max})$. This is only possible for firms with a better technology than firm j , i.e. for firms with $T_\ell > T_j$, using those notations to capture the technological component of the revenue function in both firms. The Bellman equation for the employed worker writes:

$$(\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(T_j)) V_{ij}(u) = u - c(s_i) + \delta V_i^u + \lambda(s_i, \theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{poaching}) d\Gamma(x) \quad (6)$$

where β is the discount rate, δ is the exogenous separation rate, and $\bar{\Gamma}(T_j) = \int_{T_j}^{\bar{T}} d\Gamma(x)$ is the probability of drawing a firm with a technology parameter strictly above T_j . The first term on the right-hand side is the flow utility net of search costs. The second term captures the loss from exogenous separation to unemployment. The third term is the option value of search. The following poaching utility directly stems from [Postel-Vinay and Robin \(2002\)](#):

$$u_{i\ell,j}^{poaching} = u_{ij}^{max} - \lambda(s_i, \theta_i) \int_{T_\ell}^{T_j} \frac{\bar{\Gamma}(x)}{\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(x)} dx \quad (7)$$

Optimal Search Intensity. The worker chooses s_i to maximize $V_{ij}(u)$. Taking the first-order condition with respect to s_i leads to⁷:

$$c'(s_i) = \lambda \cdot \phi(\theta_i) \int_{T_j}^{\bar{T}} [V_{ix}(u_{ix}^{poaching}) - V_{ij}(u)] d\Gamma(x) \quad (8)$$

At the optimum, the marginal cost of search effort $c'(s_i)$ equals the marginal benefit of search, i.e. the expected gain in lifetime utility from an additional unit of effort, weighted by the probability that an arriving offer comes from a firm better than j .

⁷See Appendix C.1 for the proof.

Two comparative statics follow directly from Equation 8. First, equilibrium search intensity s_i^* is increasing in θ_i : all things equal, high-productivity workers have a stronger incentive to search because a higher $\phi(\theta_i)$ raises the marginal product of search effort. Second, search intensity is decreasing in T_j and current utility u . As T_j rises toward $\bar{T}$, the integration domain $[T_j, \bar{T}]$ shrinks, so workers near the top of the job ladder search less because they are already working at one of the best firms and have little gains to expect from moving.

Job Mobility. I then derive the mobility rate of worker i , i.e. the rate at which they successfully move to a better firm. Using the job arrival function $\lambda(s_i, \theta_i)$, a move actually occurs when the arriving firm has productivity strictly above T_j , which happens with probability $\bar{\Gamma}(T_j)$. Hence, the mobility rate of worker i currently at firm j is:

$$m_{ij} = \lambda \cdot \phi(\theta_i) \cdot s_i^* \cdot \bar{\Gamma}(T_j) \quad (9)$$

Conditional on moving, hours and wages adjust following the static bargaining problem. Hours solely depend on the firm's technology T_ℓ and the worker's type θ_i , i.e. the poaching firm sets hours at the efficient level given its own technology, regardless of where the worker came from (Di Addario et al., 2023). Simultaneously, the wage must be set, by adjustment of k_j , so that the resulting flow utility equals $u_{i\ell,j}^{\text{poaching}}$. Note that the wage can thus decrease through the job change, in particular if hours move closer to the preferred level of hours at the initial level of wage.

Model Predictions. The model generates three sets of predictions. First, whether a worker is constrained depends on firm characteristics. Specifically, workers are more likely to be constrained in firms with a low α_{ij} , i.e. firms where an extra hour of work generates little additional revenue. The relation of the elasticity parameter with the productivity of the firm T_j is unclear. As T_j increases, bargained hours increase, but so do bargained wages, which may update workers' preferences towards longer hours (depending on the specification of u_i). Moreover, the profit margin p_{ij} is also likely to be correlated with T_j . Hence, the relation between hours constraints and firm's quality is ambiguous, and should be studied in the empirical section.

Second, the effect of being constrained on job mobility is also ambiguous. Constrained workers have stronger incentives to search for a better job: they have more firms to move up to, and the utility cost of working too few hours makes other jobs more valuable (Equation 8). On the other hand, they face search frictions and implicit queuing at the top of the distribution, which can offset their stronger incentives to move. All in all, the net effect of constraints on mobility is ambiguous and remains an empirical question.

Third, when a constrained worker moves to a better firm, their hours should increase, since the poaching employer sets hours efficiently given its superior technology T_j . Earnings also increase to ensure that the worker has a higher utility despite working more, but the effect on hourly wages is uncertain. Because the poaching wage is set to match the maximum utility at the incumbent firm, some of the gain in earnings from working more hours may be offset by a lower hourly wage.

3 Context and Data

In this section, I present the main aspects of the French working time regulation that are relevant for the study of hours constraints, as well as the data used throughout the paper.

3.1 French Working Time Regulation

A large range of rules and mechanisms shape French working time regulation, mostly stemming from the *Lois Aubry*, implemented between 2000 and 2002, which introduced the 35-hour workweek⁸. The standard workweek is set at 35 hours, below which part-time contracts typically operate at fixed fractions of full-time hours (60%, 80%, etc.), while a statutory maximum of 48 hours per week applies to all workers. Working hours are primarily governed by the employment contract, as hourly-paid arrangements are almost entirely absent outside a few industries. Part-time work concerns employees whose contractual hours are below the 35-hour reference duration. A key institutional feature is that part-time and full-time workers enjoy identical non-wage benefits, including healthcare and retirement rights, thereby eliminating any non-monetary incentive to prefer full-time employment.

As this paper focuses on hours constraints in France, a goal is to understand whether workers have legal ways to adjust hours in their job. The main way to do so is through overtime hours, hours worked above the contractual level, which can be performed within the limit of 220 hours per year. Overtime hours can be compensated in two different ways: either through a wage premium, as in many countries, or through additional rest (half-)days (*réduction du temps de travail*, RTT). Crucially, their completion yet solely depends on employer's decision, following the legal principle of "*employer as only judge*". As a result, workers lack any legal authority over their overtime hours.

The system of overtime hours was primarily designed to respond to temporary activity shocks, but can also be used to lengthen the workweek permanently, typically when compensated by RTT days. The 35-hour reference duration is then annualized by redistributing extra weekly hours

⁸See [Askenazy \(2013\)](#) for more details.

across the year, ensuring that the total annual hours remain equivalent to those of an employee without overtime. As introduced, the *RTT* scheme is unlikely to address the needs of constrained employees, as it reorganizes rather than increases weekly hours. However, employees do have options to convert these rest days into additional income, either by using a time savings account⁹, or by requesting direct compensation from their employer. The *RTT* scheme can thus theoretically serve as a mechanism to adjust annual earnings and hours.

The overtime system applies to both part-time and full-time workers¹⁰. Another concern is the capacity for involuntary part-time workers to move to a full-time job. The French Labor Code¹¹ states that incumbent part-time workers have to be informed about full-time vacancies in their job category by their employer and have priority to fill them. However, employers may reject such changes if they can justify that it would harm the firm's activity¹². Then, involuntary part-time work is also, to a large extent, contingent on employer-driven choices.

Finally, some employees are on a *forfait* ("package") system, which means they have established with their employer an agreement to spread out the periods for measuring working time. The most common type, *forfaits jours*¹³, is an arrangement that allow employers to count working time by days, hence not to have to monitor hours of their workers. *Forfaits jours* are distinguished by the high flexibility granted to workers in organizing their work, but come with a risk of excessive workweeks. They are therefore restricted to employees who are autonomous in organizing their schedules (e.g journalists, corporate executives) and heavily regulated. Hence, because this type of arrangement concerns a limited number of high-wage occupations and requires the employer's agreement, it cannot provide a solution to most underemployed workers.

In summary, this regulatory framework highlights that employers have important leeway in determining the number of hours of their employees, typically choosing to rely more on full-time jobs or overtime hours. On the other hand, workers do not have much margin to adjust their hours without the consent of their employer, which can lead to discrepancies between workers' preferred and actual working hours. Overall, hours constraints cannot be legitimated solely by the French working time regulation as adjustment can be made by employers. The rest of the paper examines different channels to explain why workers in practice cannot easily switch to a full-time

⁹The *Compte épargne-temps*, or time savings account, allows the employee to accumulate paid leave entitlements or to receive remuneration (immediate or deferred), in return for leave or rest periods not taken. The establishment of this system in the firm is yet non mandatory.

¹⁰The terms *heures complémentaires* and *heures supplémentaires* are respectively used for part-time and full-time workers to denote overtime hours.

¹¹See Article L3123-3.

¹²Following a dispute in October 2023, the *Cour de Cassation*, the French supreme court for civil and criminal cases, reinforced the claim that employers must be able to explain their decision.

¹³In 2024, 17% of salaried workers were working under a *forfait jours* agreement. See [Breda et al. \(2025\)](#) for more on this topic.

job or work overtime hours.

3.2 Data

I mobilize for the period between 2003 and 2023 two classic data sources for the study of the labor market in France: the French Labour Force Survey, the *EEC*, and the French matched employer-employee dataset, the *DADS*¹⁴. Furthermore, both sources are used to build a newly matched dataset, the $EEC \times DADS$ which combines their information at the individual level. Beyond its purpose for this paper, this unique linkage constitutes an important contribution to the integration of survey and administrative datasets.

3.2.1 French Labour Force Survey (EEC)

The *Enquête Emploi en Continu* (EEC) is a large-scale nationally representative survey conducted by the French National Institute of Statistics and Economic Studies (*Insee*). Like every other European Labour Force Survey, it complies with the guidelines of the International Labour Organization (ILO) and of Eurostat. Designed to provide a thorough understanding of the labor market dynamics, the EEC collects detailed information on workers' characteristics at a quarterly frequency for a maximum period of 6 quarters. Reported variables include standard demographic and labor market indicators such as age, municipality of residence, occupation, and labor earnings, as well as a broad range of measures capturing worker preferences. The occupation classification system underwent two revisions during the observation period, in 2008 and 2020. I harmonize occupation codes throughout the entire period using correspondence tables provided by Insee¹⁵. The EEC includes two measures of hours worked, the central outcome of this paper, reported usual hours worked and the number of hours worked during the reference week in which the worker is surveyed, which I do not use in the main text of the paper¹⁶. The EEC sample consists of 112,523 part-time workers and 512,944 full-time workers, after restricting to non-agricultural salaried workers between 18 and 64 years old with non-missing hours worked and an hourly wage between 0.8 and 1000 times the hourly minimum wage. Most importantly, the survey contains specific variables that I use to provide direct measures of preferences on hours worked.

Hours preferences are measured using binary responses to questions STPLC and STMN, respectively asking workers whether they "*want to work [more/less] hours with a corresponding [increase/decrease] in earnings*". Workers who report wanting to work more or less are respectively denoted as con-

¹⁴To be consistent with previous work, e.g. [Godechot et al. \(2025\)](#) or [Bergeaud et al. \(2025\)](#), I refer to this data as *DADS* with a slight abuse of notation, as its official name is *Base Tous Salariés* (BTS).

¹⁵When the correspondence tables do not provide a perfect 1-to-1 match at the 4-digit level (497 values), I assign observations to the most common occupation code in the new classification.

¹⁶Usual hours worked are only used in Appendix [B1](#) and [B2](#) for robustness checks over the validity of variables on worker preferences and paid hours worked.

strained *from above* or *from below*, following the terminology of [Lachowska et al. \(2026\)](#). Workers who answer no to both questions are referred to as *unconstrained*. The specification “*with a corresponding income variation*” is designed to elicit workers’ optimal hours at their current hourly wage, consistent with traditional labor supply models¹⁷. The nonresponse rate is less than 1%, making this variable well-suited for analysis. When individuals respond affirmatively to either question, they are subsequently asked “*the number of hours that they would ideally work with a corresponding income variation*”. This follow-up question yields a measure of desired hours for constrained workers, providing an intensive margin component to the measure of hours constraints. In the group of workers constrained from above, I define as *involuntary part-time* workers the individuals who work part-time and report desired hours per week superior or equal to 35, the full-time frontier. More than 70% of part-time workers constrained from above are in this case.

Importantly, the wording of the question associated with the variable STPLC, on which much of the empirical work is based, underwent one substantive change. From 2003 to 2012, the question was framed with reference to the specific survey reference week, asking whether the respondent wished to work more hours *during that particular week*. Starting in January 2013, following a comprehensive questionnaire redesign, the question was broadened to ask whether the respondent would like to work more hours with a corresponding increase in income without any temporal anchor. According to INSEE’s own documentation ([INSEE, 2014](#)), this change in framing raised the measured stock of underemployed workers by approximately 200,000 persons, constituting a well-documented series break. A further restructuring of the questionnaire took place in 2021 to comply with the new European regulation, which reorganised question modules and introduced internet self-administration, but did not alter the conceptual scope of STPLC.

The presence of a break in 2013 is addressed in two complementary ways. First, all regressions include year fixed effects, which absorb any level shift in STPLC that is common across workers in a given year. Second, as a robustness check in Section 4.1, I apply a multiplicative correction that anchors on the post-2013 methodology and rescales the 2003–2012 segment so that December 2012 and January 2013 are level-consistent (Figure A2). This correction allows to represent the long-run descriptive picture of hours constraints.

Similar questions appear in other Labour Force Surveys, and have been frequently used to measure hours preferences in previous research (see e.g. [Gaini and Vicard \(2012\)](#); [Beckmannschagen and Schröder \(2022\)](#); [Asai \(2024\)](#) or more recently [Jarosch et al. \(2025\)](#)). However, the reliability of the answers can be questioned as common survey biases apply here (see [Stantcheva \(2023\)](#)). Understanding how workers perceive questions regarding the distance to their ideal hours, and whether their answers are truly indicative of constraints, is a priori challenging. Validity concerns

¹⁷See [Jarosch et al. \(2025\)](#) for a more thorough discussion of that point.

are addressed to a certain extent by exploiting other survey information from the EEC. For instance, 95% of workers who report being constrained from above report their availability to work more, the share being slightly higher for part-time workers. Also, 89% of them want to adjust their hours in their current primary job and, in the case of part-time workers, are less prone to look for a secondary job. This provides evidence that answers do not reflect mere preferences that are not in regard to real-life constraints (e.g. child caring). These workers have internalized their work environment and are ready to work more, preferably in their current main job. Appendix B1 presents additional evidence on work motivations, persistence of preferences and worker mobility over the 6 quarters of observation to discuss the interpretation and reliability of this variable. It also includes descriptive statistics regarding the distribution of desired hours in the sample.

Importantly, I use a version of the French Labour Force Survey that provides access to the establishment's identifier (the *SIRET*) for each worker's place of employment, as coded by Insee based on information from the worker's payslip. This feature, which to my knowledge is unique to the French version of the survey, is crucial to this study. The remainder of the paper considers only individuals for whom the *SIRET* information is available — 82% of the original sample. This source of data enables the study of hours constraints reported by each worker while accounting for firm-specific effects, a contribution that lies at the heart of this paper.

3.2.2 French Matched Employer-Employee Dataset (DADS)

The second source of data is the French matched employer-employee dataset *Base Tous Salariés* based on social security records (the *Déclarations Annuelles de Données Sociales / Déclaration Sociale Nominative*) and hereafter called *DADS*, notoriously used in Abowd et al. (1999). This dataset is constructed by Insee at an annual frequency, based on administrative employer reports (mandatory for each employee subject to French payroll taxes) containing information on characteristics of the worker. In particular, it includes their income, paid hours worked, and length of employment spell. I use the *Fichiers Postes* version of the *Base Tous Salariés*, an exhaustive version that covers all jobs in the entire salaried workforce in each calendar year. The chaining procedure described by Godechot et al. (2025) is then used to build a quasi-exhaustive *DADS* panel dataset over the period between 2003 and 2023, which enables the observation of employer-to-employer transitions for almost the entire workforce.

Using employer-reported paid hours worked as a measure of actual hours worked raises methodological concerns. Paid hours, as reported in the *DADS*, include all remunerated periods, i.e. not only regular working weeks but also paid leave. Hence, this definition tends to reflect contract hours and may a priori not be able to capture small adjustments. Appendix B2 assesses the quality of hours worked data by reproducing the approach of Lachowska et al. (2022) on Washington

state administrative data, augmented with features specific to the French labor regulation and the data I use¹⁸. The implemented tests suggest that the quality of the data is comparable to the data used in [Lachowska et al. \(2022\)](#). As expected, paid hours tend to be quite concentrated around the reference duration (about 30% of the population works the equivalent of 35 hours per week) but there remains substantial variation likely driven by part-time work and overtime hours.

A particular feature of French working time regulations, presented in Section 3.1, requires special consideration. *Forfait jours* contracts, covering approximately 17% of the workforce in 2024, measure working time in annual days rather than weekly hours. Under this arrangement, employers are not required to monitor or report actual hours worked in DADS records¹⁹. Insee addresses this data gap by imputing standardized annual hours (typically 1,820 or 2,200 hours depending on the year) of these workers, identifiable through the UNITMESUREREF variable since 2017. Given that these imputed values do not reflect actual working time and would introduce systematic measurement error into the analysis, workers in occupations with a large share of *forfait jours* contracts are excluded from the sample throughout most of the paper, with exclusions explicitly noted. This restriction barely affects the sample of, primarily low-skilled, constrained workers as this type of contracts is only available to high-skill workers with strong autonomy in their job, e.g. journalists or consultants.

3.2.3 EEC $\times$ DADS Sample

Finally, I adopt a fuzzy matching approach to link together the EEC and DADS datasets and build a unique new dataset. Given the quasi-exhaustive nature of the DADS data on employment spells, most individuals surveyed in the EEC should appear in the DADS. In addition, workers should be identifiable based on information, supposedly consistent across sources, on age, sex, occupation, part-time/full-time work, birth department, birth month (only between 2009 and 2012), residence municipality, and, crucially, establishment's ID. This information is thus mobilized to build a correspondence table between individual identifiers of both datasets. The procedure works as a "1-to-1"²⁰ matching algorithm on common variables, i.e. workers surveyed in the EEC are identified in the administrative records based on precise identical information across sources. This linkage is made possible by the presence in the EEC of the worker's establishment's ID (the *SIRET*). Impor-

¹⁸In particular, the linkage of EEC and DADS datasets allows me to compare employer-reported and employee-reported hours for the same worker. Such comparisons have already appeared in previous research by [Frazis and Stewart \(2010\)](#). This work yet stands out as I rely on a much larger sample and on an administrative source of information on the employer's side.

¹⁹[Breda et al. \(2025\)](#) provides more background on this type of contracts and addresses the implications of workers' misperceptions about them on working conditions.

²⁰I refer to a "1-to-1" match in this context to describe a situation where exactly one worker in the DADS dataset corresponds to exactly one worker in the EEC dataset based on identical values for the specified matching variables. Likewise, a "1-to-many" match depicts a situation where multiple workers in the DADS dataset corresponds to a unique worker in the EEC.

tantly, the method used here relies on fuzzy matching, and thus can theoretically lead to errors in the linkage of IDs. However, I reckon that the high precision of variables used and the conservative decision to remove all "1-to-many"²⁰ matches ensure a reliable linked dataset. The linkage is performed for every year between 2003 and 2023, keeping all quarterly information from the EEC and yearly information on all employment spells from the DADS. Appendix B3 details the entire matching procedure.

The operation yields a sample of 435,472 perfectly matched individuals between 2003 and 2023, a coverage of 67% of the EEC sample of employees²¹ on this period. Cases where individuals cannot be matched across sources occur for three main reasons: (a) the establishment's ID is missing or incorrectly reported by the worker, (b) individuals are dropped during the cleaning process of the DADS as their information on hours or earnings is missing, (c) multiple workers cannot be distinguished based on the matching variables. The quarterly nature of the EEC and the linkage of all DADS employment spells (including secondary jobs) over the period imply multiple observations per worker and year, hence generating a sample of 1,695,359 observations. This sample, hereafter denoted as the EEC-DADS sample, shows reasonably representative as described by summary statistics in Table A1. The most striking disparity with respect to the original EEC sample is the sizable under-representation of constrained part-time workers (representing 26% of part-time workers in the EEC-DADS sample against 34% in the original). This likely occurs because this population tends to hold more precarious positions with lower compliance on administrative duties, thus affecting the reliability of their information in both sources. As a result, the hours constraints that affect the part-time population in the EEC-DADS sample should reflect more stable and structural constraints than those in the original EEC sample. This is consistent with validity checks in the data: part-time workers who wish to increase their hours hold a permanent job more frequently (72% against 60% in the original sample) and have more experience in their current firm (76% have more than 1 year against 70% in the original sample).

Then, 92% of the workers in the EEC-DADS sample are connected, using their yearly DADS identifier, to the previously chained DADS panel between 2003 and 2023. This dataset is denoted as the *EEC-DADS Panel*. As discussed in Godechot et al. (2025), the DADS panel is not fully exhaustive, which explains why some workers of the EEC-DADS sample cannot be recovered. Their chaining procedure is based on the presence for each individual of information on the current and the previous year of observation, so that workers who are not employed for one calendar year cannot be chained. Hence, the procedure is likely not able to capture very precarious workers who jump in and out of employment, but rather focuses on relatively stable salaried workers. The EEC-DADS Panel combines for each individual EEC-based information over 6 quarters of observation, here-

²¹ After removing observations with missing number of hours worked or missing establishment's ID.

after the *EEC period*, and DADS-based information over the period of appearance in the panel. In other words, it contains information on individual preferences in terms of hours over a short period of time and on their employment history over a long period. The particular structure of this dataset is illustrated in Figure A1. The EEC-DADS Panel gathers information on 399,553 individuals appearing 10 years on average in the panel, sometimes with multiple spells of employment over one year, resulting in a total of 9,734,018 observations.

This unique dataset paves the way for new empirical work on the topic of hours constraints by combining a self-reported measure from the Labour Force Survey with the panel dimension and reliability of administrative employer-employee datasets. Beyond this study, this dataset has potential applications across a wide range of research purposes, as the Labour Force Survey includes extensive information on various topics.

4 Sorting of Constrained Workers

This section provides empirical evidence on the sorting of constrained workers by exploiting an individual survey-based measure linked with employer information. Section 4.1 presents the main characteristics of workers who report being constrained in their hours, as well as their distribution across occupations and industries. Section 4.2 examines the role of firm sorting in shaping hours constraints. Section 4.3 tackles the remaining variation in hours between constrained workers and workers from the same workplace.

4.1 Descriptive Statistics

As a first step to the empirical part, I examine the characteristics of workers who report desired hours above their current level. As presented in Section 3.2, three groups of workers are distinguished based on their answer to questions regarding hours preferences in the Labour Force Survey: workers *constrained from above* who report that they ideally want to work more hours at their given wage rate, workers *constrained from below* who report that they ideally want to work less hours at their given wage rate and *unconstrained* workers who report no will to work different hours at their given wage rate. The defined groups respectively represent 19%, 2.5% and 78% of the Labour Force Survey sample between 2003 and 2023. This indicates that nearly one out of five salaried workers is constrained from above, while the share of workers constrained from below is minimal²². For this reason, the term *constrained* is used in the rest of the paper to refer to the former situation.

²²This stands in sharp contrast with the prevalence of overwork in the German case, as documented by Jarosch et al. (2025). As mentioned in Cohen et al. (2025), the distributions of desired hours are similar across countries but actual hours are higher in Germany, hence the difference in the direction of constraints.

The French working time regulation splits employees in two groups based on their working hours. Workers who are hired in a contract with strictly less than 35 hours per week are part-time workers, while others are full-time workers. This distinction between both types of workers is conserved throughout the paper as the profile of hours preferences differs in both groups. Part-time workers are largely over-represented in the constrained population: 34% of part-time workers report wanting to increase their hours, against 17% in the full-time group (see Table A1). In addition, about 70% of constrained part-time workers are involuntary part-time workers, i.e. they would ideally work as full-time employees, as derived from their desired hours²³.

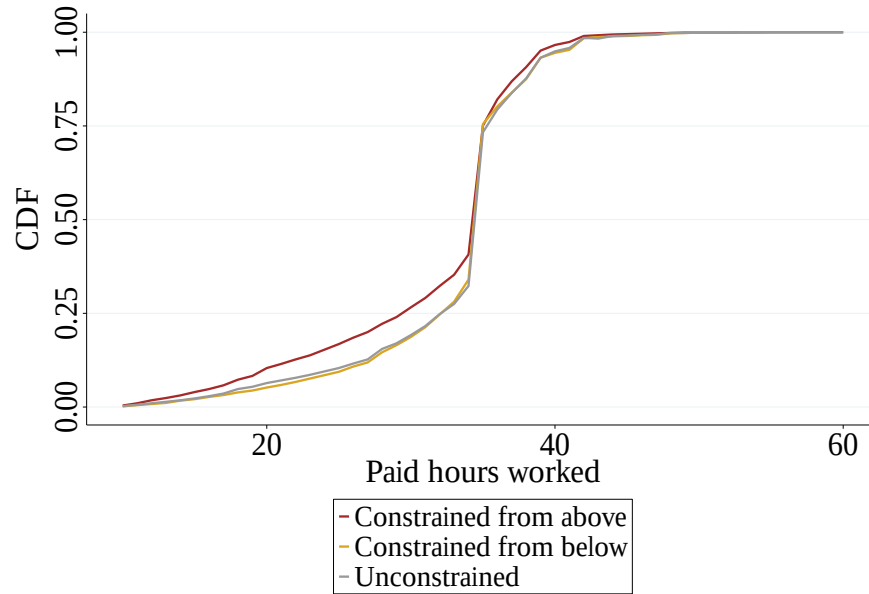
Figure 1 displays the cumulative distribution functions of paid hours worked by group of hours preferences. The sample used is the matched EEC-DADS sample in order to use paid hours from the administrative data rather than reported usual hours worked. Workers who want to increase their hours work significantly fewer hours than others, essentially because of the over-representation of part-time workers. In the full-time zone, the three curves are very close, with constrained-from-above workers working only about 0.1 hour per week less than other groups. It means that constrained full-time workers do not significantly differ from other full-time workers in terms of hours worked. The figure thus acknowledges a difference in the interpretation of preferences across both types of workers. Part-time workers who want more hours primarily aim to secure a full-time job, the standard employment status in the labor market, to ensure decent conditions of living. On the other hand, full-time constrained workers already have this status and tend to work about the same number of hours as any other worker. Their report thus relates more to particular hours preferences than to being constrained to short-hour jobs. Self-reported underemployment is therefore a weaker signal for full-time workers, and the rest of the paper primarily focuses on the case of part-time constrained workers.

Figures 2a and 2b illustrate how the shares of constrained workers vary in the (hourly) wage distribution, separately for part-time and full-time workers. Both figures show that hours constraints are sizable and that the probability to want more hours is regressive in wages, particularly in the part-time case. At the bottom of the distribution, about half of part-time workers are constrained. A simple interpretation is that some workers aim to increase their earnings through hours worked to compensate for low wages (consistently with Figure B1b), i.e. a cross-sectional “income effect” is at work. Conversely, the probability to be constrained from below is positively correlated with wages, though it remains marginal even at the top of the wage distribution.

Table A3 presents further details about the composition of each group, although it is well summarized by the relation with wages. Determinants of being constrained from above include being young, low-skilled, born outside of France, employed in a private sector firm, employed in a small

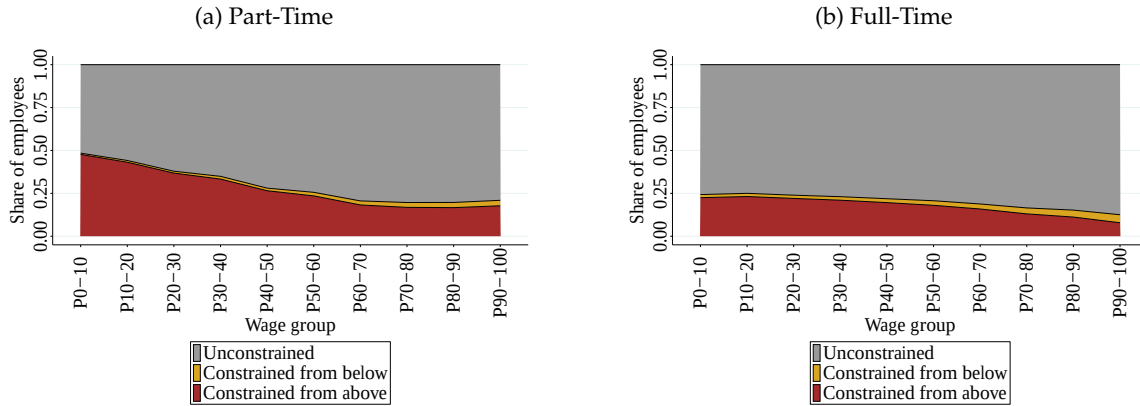
²³This corresponds to the PTWFT category in previous work on underemployment (Bell and Blanchflower, 2021).

Figure 1: Hours Worked by Group of Worker Preferences



Note: The figure shows the cumulative distribution functions of paid hours worked by group of worker preferences in the EEC-DADS sample. Three groups are considered based on their answers to the following questions in the surveys: "do you want to work [more/less] hours in your job with a corresponding income variation?". "Constrained from above" workers report that they want to work more hours; "Constrained from below" workers report that they want to work less hours; "Unconstrained" workers answer no to both questions. Hours worked are paid hours from the DADS recomputed as full-year equivalent.

Figure 2: Worker Preferences in Hours in the Wage Distribution



Note: The figures show the distributions of worker preferences based on Labour Force Surveys pooled between 2003 and 2023. Three groups are considered based on their answers to the following questions in the surveys: "do you want to work [more/less] hours in your job with a corresponding income variation?". "Constrained from above" workers report that they want to work more hours; "Constrained from below" workers report that they want to work less hours; "Unconstrained" workers answer no to both questions. Wage deciles are defined within year and employment status (part-time/full-time).

firm and in a non-permanent contract. The subgroup of involuntary part-time workers is very different from the rest of constrained workers. For instance, women, mostly working in low-skill tertiary jobs, represent a large majority (77%) of this group. Note that secondary employment

remains marginal even among involuntary part-time workers, which suggests that constrained workers do not accept a part-time job in order to combine it with a secondary job.

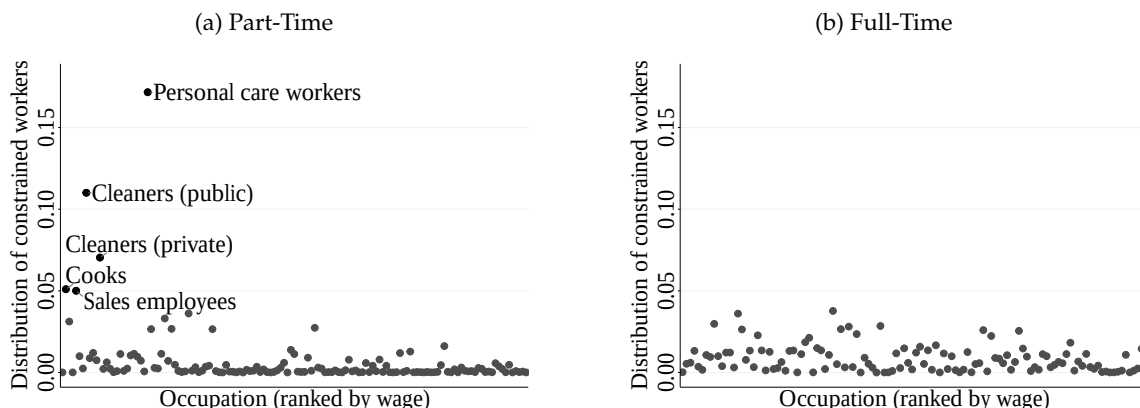
A natural concern when studying hours constraints is whether their prevalence reflects structural features of the labor market or simply tracks business cycles. In this regard, [Borowczyk-Martins and Lalé \(2020\)](#) show in the US context that involuntary part-time employment is strongly countercyclical, reflecting that some employers tend to adjust their workers' hours downward rather than destroy jobs during recessions. To assess this, Figure [A2](#) plots the quarterly share of workers reporting that they want to work more hours over the 2003–2023 period. Before doing so, however, I need to address a methodological break in the French Labour Force Survey that generates an artificial discontinuity in the raw series. As explained in Section [3.2.1](#), a change in the questionnaire wording introduced in January 2013 produces a sharp upward jump in the measured share of constrained workers that is inconsistent with any plausible labor market development. To recover a comparable series over the full period, I apply a multiplicative correction anchored on the post-2013 methodology: the 2003–2012 segment is rescaled so that its level in December 2012 matches that of January 2013 using the survey reference week date. The corrected series reveals a broadly stable share of constrained workers throughout the period, with no clear cyclical pattern. If anything, the corrected series suggests a mildly procyclical pattern, with the share of constrained workers reaching its peak during the relatively high-growth years of 2003–2007. A simple regression of the annual share of constrained workers on GDP growth yields a positive but statistically insignificant coefficient. The evidence is thus more consistent with hours constraints being a structural feature of the French labor market than a cyclical one.

To end this section, I focus on the relation between occupations and hours constraints. Prior research ([Altonji and Paxson, 1986](#)) has often considered that working hours are largely occupation-specific. For instance, assembly line workers exhibit strong uniformity in hours worked due to standardized shift schedules (such as 8-hour shifts) common in manufacturing contexts. Therefore, it seems reasonable that some workers face constraints on their hours because their jobs structurally involve shorter working periods. To address this point, I leverage information on hours constraints from the EEC at the occupation level.

Figures [3a](#) and [3b](#) represent the probability density function (PDF) of constrained workers, split by working time status, across occupations at the 3-digit level (110 values). Each dot represents one occupation and the y-axis corresponds to the share of all constrained workers who work in that occupation. First, part-time workers who want more hours are heavily concentrated in low-wage jobs. In particular, five occupation groups (personal care workers, cleaners in the public and private sector, sales employees, cooks and waiters) account for half of constrained part-time workers. On the other hand, constrained full-time workers are relatively uniformly distributed across

occupations with no clear concentration in low-wage jobs. Occupations with a strong concentration of constrained part-time workers are notorious for their relatively high reliance on part-time employment, but still employ a non-marginal proportion of full-time workers. Figure A3 plots the cumulative density function (CDF) of paid hours in each of the five main occupations for constrained part-time workers, with the vertical red line representing the frontier between part-time and full-time work. While only 16% of personal care workers are employed at full-time, the shares of part-time and full-time work are balanced in other occupations. In addition, using the DADS data and a simple decomposition, I find that 81% of the variance in paid hours remains within occupations at the 3-digit level. This shows that schedule rigidities specific to certain tasks cannot explain the degree of hours constraints, in particular involuntary part-time work, in most occupations.

Figure 3: Distribution of Constrained Workers by Occupation



Note: The figures show the distributions of constrained (from above) workers across occupations for part-time and full-time workers. Data is based on Labour Force Surveys pooled between 2003 and 2023. Each dot corresponds to an occupation (ranked by average wage) at the 3-digit level in the PCS classification (110 values). Average wages are computed from the DADS data between 2009 and 2019, separately for part-time and full-time workers. The y-axis indicates the proportion of the population constrained from above that works in a given occupation.

This section establishes that workers who want to increase their hours work on average less hours than other workers. This gap is almost entirely driven by the higher proportion of part-timers, as average hours of full-time workers are broadly similar across groups of preferences. For full-time workers, to want more hours then mostly reflects idiosyncratic preferences rather than a particularly low level of hours, i.e. constrained workers are, by and large, workers with regular hours who want to work above-average hours. The situation is different for constrained part-time workers, as most of them want a regular full-time job, the benchmark of the labor market, but cannot access them. Underemployment thus carries a different interpretation in both groups. The

section also shows that constrained part-time workers are concentrated in a small number of low-wage occupations, but that the latter occupations have, at least, a decent share of full-time workers. This suggests that hours constraints do not fully coincide with occupation-specific requirements but rather depict a segmentation between workers doing the same job. The next section examines whether firm sorting captures this segmentation.

4.2 Firm Sorting

This section studies the role of firm-specific effects in the phenomenon of hours constraints. I take advantage of the presence of establishments' IDs in the French Labour Force Survey to examine the distribution of constrained workers across firms. This approach complements recent work by [Labanca and Pozzoli \(2023\)](#) on the role of firms in hours constraints by using survey-based evidence of underemployment. Here, I also connect worker hour preferences with employer hour policies, a relation at the heart of [Lachowska et al. \(2026\)](#)'s work, by which this section is largely inspired. The section further provides evidence on the interaction between firm effects and occupational effects, i.e. the extent to which firm heterogeneity explains variation in hours within occupations.

To capture firm-specific preferences for hours, I use firm effects estimated from the AKM specification ([Abowd et al., 1999](#)), using hours worked as the dependent variable. These effects provide a measure of a firm's premium in hours and are commonly interpreted as employer hour policies ([Lachowska et al., 2026](#)). Using a separate AKM specification, I also estimate firm effects on hourly wages, the original dependent variable used by [Abowd et al. \(1999\)](#). Firm wage effects have been shown to be strongly correlated with firm productivity measures, with more productive firms paying higher wages ([Card et al., 2013](#); [Carlsson et al., 2016](#)). In the following, I connect hours constraints to these two types of firm-specific features.

AKM Estimation. As in [Babet and Chabaud \(2024\)](#), I use a chained DADS panel between 2005 and 2019, quasi-exhaustive of the French salaried workforce, and apply several restrictions to the sample²⁴. Hours worked are annualized paid hours, and wages correspond to real net hourly

²⁴Workers aged between 18 and 64, in ordinary jobs of more than 120 hours and 60 days, with hourly wage greater than 0.8 minimum wage and smaller than 1,000 minimum wages. I also keep only one observation per person-year so I take the annual dominant employer of the worker (defined as the employer from which the worker earns the most during the year). Occupations with more than 20% of *forfait jours* workers in the EEC between 2013 and 2023 are excluded from the sample.

wages. The sample is also restricted to the largest weakly connected set of firms, as is common in this literature (Card et al., 2013). The estimation sample contains 29,824,763 workers and 1,318,726 firms, and covers 92% of firms that appear in the EEC-DADS sample between 2005 and 2019²⁵. The following model is fitted:

$$y_{it} = \phi_i + \psi_{j(i,t)} + X_{it}\beta + u_{it}$$

with y_{it} the logarithm of the outcome (either hours or wages) of worker i in year t . ϕ_i is the fixed effect of worker i , and $\psi_{j(i,t)}$ is the fixed effect of firm j , the employer of worker i during year t . Time-varying covariates X_{it} are limited to fixed effects for years and age as a cubic polynomial. u_{it} is the idiosyncratic error term. This model can be used to decompose the variance in hours and wages into components associated with worker and firm heterogeneity and sorting, as done by Babet and Chabaud (2024) with the same data²⁶. The goal here is rather to use the estimated firm effects for their interpretative value. Figures A4a and A4b show an event study check popularized by Card et al. (2013) to assess the plausibility of exogenous mobility in the sample, respectively for hours and wages. The symmetry of outcome evolutions suggest that the specification with fixed worker and firm effects rationalizes well employer-to-employer moves in the data.

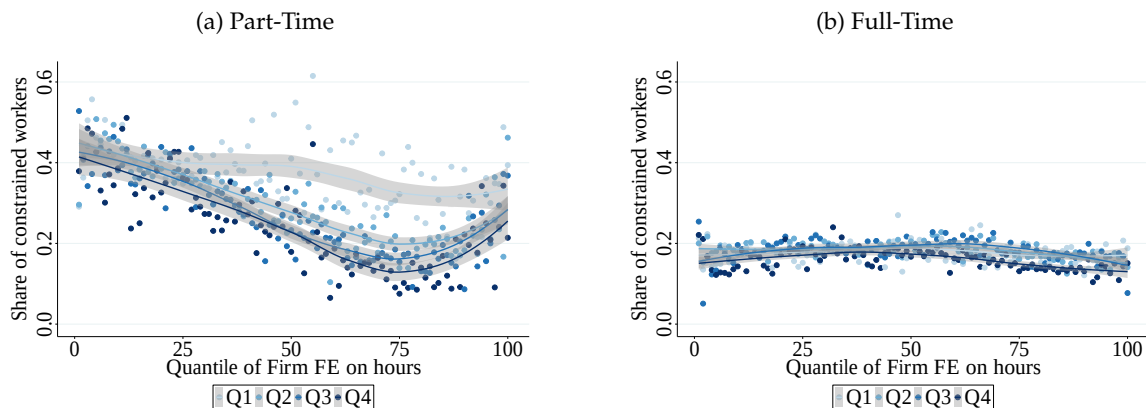
Firm effects are split into percentiles for hours and into quartiles for wages, and each combination of hour-effect percentile and wage-effect quartile is associated with the share of workers within the cell who report being constrained from above. Each such combination thus characterizes groups of firms with similar joint policies on hours and wages. Figures 4a and 4b illustrate the relation between worker preferences and firm policies, respectively for part-time and full-time workers, and confirm differences between both types documented in previous sections. Two features stand out from Figure 4a, which covers part-time workers. First, the share of constrained workers is highest among firms with both low-hours and low-wage policies, and declines steadily as firms offer more hours, with a moderate upturn at the very top of the hours ranking. Second, the curve for the lowest wage quartile sits well above the others and remains relatively flat across the hours distribution, indicating that constrained workers are systematically over-represented in low-paying firms regardless of their hours policies. Figure 4b presents a more uniform picture for full-time workers, as the share of constrained workers is particularly stable across all groups of hours and

²⁵Uncovered firms are firms of less than 10 employees withdrawn of the sample by the connected set restriction. Despite the over-representation of constrained workers in small firms, as shown in Table A3, the estimation allows the retrieve the AKM firm effect associated with 95% of constrained workers.

²⁶They find that 15% of the variance of the (log) number of hours worked can be attributed to firm effects.

wage firm effects, with all curves flat and confounding each other at 20%, the share of full-time constrained workers in the entire EEC sample.

Figure 4: Hours Constraints and Firm Policies



Note: The figures represent the relationship between hours constraints and AKM firm effects on hours and hourly wages for part-time and full-time workers. The x-axis represents the percentile of firm fixed effect on hours, while the y-axis shows the share of workers constrained in their hours. Groups (Q1-Q4) represent different quartiles of firm effects in wages, with darker points indicating higher wage quartiles. All firm effects are estimated through an AKM decomposition. The data used is a panel based on chained DADS yearly files between 2003 and 2023. Hours worked are paid hours recomputed as full-year equivalent. Curves represent the fit of non-parametric local regressions (LOESS). Shaded areas correspond to 95% heteroskedasticity-robust confidence intervals.

The result for part-time workers is consistent with the existence of a job ladder across firms, previously introduced in [Lachowska et al. \(2026\)](#) as a mechanism for hours constraints. Most constrained workers are employed in low-quality firms, with low hours and wage policies, and seek to climb the ladder to improve their earnings. Yet, the limited availability of such jobs implies that some part-time workers end up constrained. This is consistent with the theoretical framework developed in Section 2, considering that the marginal productivity of an additional hour of work reflects firm quality. By combining individual-level survey information on constraints with firm effects on hours, this figure provides prominent empirical evidence for this mechanism in the case of hours constraints. No such pattern is observed in the case of full-time workers, which I interpret as additional evidence that constrained full-time workers are workers in regular jobs with particular preferences.

This section highlights the role of firm sorting in the fact that some individuals are constrained in their hours while others are not. Using a simple ANOVA-style variance decomposition, I find that 52% of the total variance in hours worked is captured by the intersection of occupation (at the 3-digit level) and firm effects. Then, while constrained workers cannot work long hours partly

because of their jobs' requirements, there often exists workers with a similar profile who get to work more hours.

4.3 Local Variation in Hours Worked

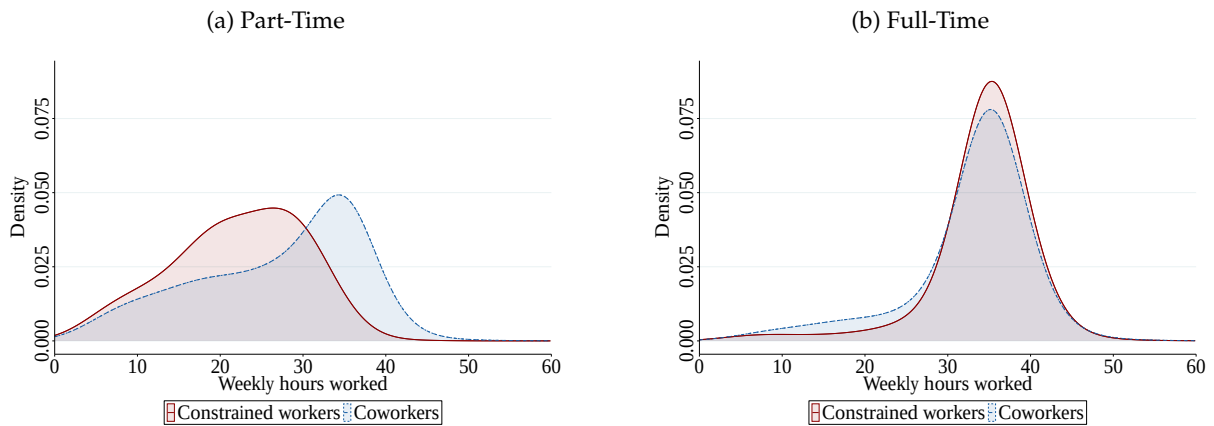
Previous sections have documented the sorting of constrained workers across occupations and firms, suggesting that hours constraints partly reflect features specific to certain tasks and employers that structurally limit hours worked. This section shows that a non-negligible variation in hours remains at a *local* level, typically for a given job in a given establishment. Using the EEC-DADS dataset, I can combine administrative information on constrained workers and on other workers from the same firm. This allows me to observe where constrained workers stand in their local hours distribution and to investigate mechanisms that can explain their relative position to other workers performing the same job.

I consider workers in the DADS exhaustive records who work in the same establishment (*SIRET*) and occupation as a worker identified as an involuntary part-time worker in the EEC. Given the exhaustive nature of this data, this definition should gather all simultaneous coworkers of involuntary part-time workers from the EEC sample. Note that the methodology does not capture coworkers' preferences but instead provides a comprehensive view of the hours distribution surrounding constrained workers. The goal is to understand why some workers work more hours, or have a full-time job, while similar workers are unable to do so. Occupations are defined at the 2-digit level (28 values), which, in a given workplace, ensures a comparison to similar workers. I consider the 2-digit rather than the more precise 3-digit level, in order to include all years between 2003 and 2023, across which the PCS classification is not consistent at the 3-digit level. I apply filters to the data similar to [Babet and Chabaud \(2024\)](#) to ensure the relative stability of coworkers in their job²⁷. After restrictions, the dataset is composed of 11,689,428 observations of coworkers. Out of the 124,916 constrained workers in the EEC-DADS sample, 101,622 (81%) have at least one worker in the same establishment $\times$ occupation $\times$ year, including 16,202 involuntary part-time workers. The sample of constrained workers here is implicitly restricted to typical salaried workers who work in a workplace with other workers. Certain occupations with strong hours constraints, e.g. personal care workers, are likely not captured in this section.

²⁷Workers are aged between 18 and 64, have a stable working contract (*CDD* or *CDI*) lasting at least 60 days, with annual hours above 120, and hourly wage between 0.8 and 1,000 times the minimum wage.

Using the triplet establishment $\times$ occupation $\times$ year as the definition of a job, a simple variance decomposition yields that 42% of the total variance in hours remains within-job. Figures 5a and 5b plot the kernel density estimates of weekly hours worked separately for constrained workers and their coworkers, for part-time and full-time workers respectively. To account for the unequal number of coworkers across firms, the coworker distribution is constructed using inverse group-size weights, so that each constrained worker's coworker group contributes equally to the coworker distribution. For involuntary part-time workers, the distribution of hours is shifted to the left relative to that of their coworkers, indicating that constrained workers tend to work fewer hours even within the same job. In addition, 85% of involuntary part-time workers share their job with at least one full-time worker, and in those jobs the average share of full-time workers is 55%. Involuntary part-time work thus does not stem from an absence of full-time work within the firm, but instead relies on worker-level differences in hours worked. Full-time constrained workers tell a different story. By construction, their hours distribution sits above that of their coworkers, since this group includes part-time workers. When restricting the comparison to full-time coworkers, there is no significant difference between the two curves. The full-time hours distribution is also more compressed, e.g. 6% of full-time coworkers work more than 40 hours per week²⁸. Hence, there is limited scope for hours increases within full-time jobs without a change of job or occupation.

Figure 5: Distribution of Hours Worked by Constrained Workers and Coworkers



Note: The figures plots kernel density estimates of weekly hours worked for constrained workers (solid line) and their coworkers (dashed line). Panel (a) displays part-time constrained workers; panel (b) displays full-time constrained workers. The coworker distribution is constructed using inverse group-size weights, whereby each coworker is assigned a weight equal to the inverse of the number of coworkers associated with the same constrained worker, ensuring that every constrained worker's peer group contributes equally to the distribution.

²⁸This threshold is arbitrarily chosen as it is the most common target reported by full-time constrained workers.

The evidence from Figure 5a points to a form of within-firm worker segmentation, whereby part-time and full-time workers coexist in the same jobs, yet some workers remain stuck in part-time arrangements. This raises the question of why certain workers fail to secure a full-time job when such jobs demonstrably exist within their firm. The theoretical framework introduced in Section 2 offers a rationalization of this within-firm variation in hours based on heterogeneity in worker productivity as perceived by the firm. Under this view, workers who work longer hours are regarded by the firm as sufficiently productive to justify additional costs, so that hours become a signal of the firm’s assessment of worker quality. Hours constraints would then reflect not an economy-wide shortage of full-time work, but rather a firm-level mechanism that divides hours unequally between workers of the same establishment.

To investigate the mechanisms behind within-job variation in hours, I compare the characteristics of involuntary part-time workers with those of their full-time coworkers. The DADS data includes information on gender, age, type of contract (permanent or non-permanent), native or foreigner status (derived from the birth department variable), distance to workplace (based on residence and workplace municipalities²⁹), and tenure in the firm (derived from the DADS Panel)³⁰. Age is split as a set of five categories to allow for non-linearities, and tenure is winsorized at five years.

Table 1 reports the estimated differences in characteristics between involuntary part-time and full-time workers in the same job. Each coefficient is obtained from a separate regression controlling for all other characteristics. Involuntary part-time workers differ in several dimensions. The age profile follows a clear pattern: involuntary part-time workers are significantly more likely to be under 45, with a positive and significant relationship for the 15-24, 25-34, and 35-44 age brackets, while the relationship turns negative and significant for workers aged 45-54 and 55+. This result should be interpreted with caution given the over-representation of older workers in the EEC-DADS sample (see Table A1). Gender plays a more reliable role: being an involuntary part-time worker rather than a full-time worker is associated with a significant 14 pp higher probability of being a woman. Tenure exhibits a small negative and significant correlation with hours constraints, suggesting that workers are more likely to obtain a full-time contract after a few years in the same firm. The “Foreigner” dummy is positively and significantly associated with involun-

²⁹Distances between municipalities are calculated using the Haversine formula, which computes the great-circle distance between two points on a sphere given their latitude and longitude coordinates.

³⁰For this exercise, the sample of constrained workers is restricted to years between 2008 and 2023, in order to measure tenure in the firm for at least 5 years before.

tary part-time work, indicating that foreign-born workers are somewhat more likely to be hours-constrained than their native-born coworkers in the same job, conditional on the other characteristics in the regression. Distance to commute is negatively and significantly associated with involuntary part-time status, with involuntary part-time workers living on average 5.2 km closer to their workplace. Finally, holding a permanent contract is negatively and significantly correlated with involuntary part-time work, consistent with hours constraints being more prevalent among workers on non-permanent contracts. Considering results on age and tenure, this section could hint at involuntary part-time work as a labor market entry phenomenon. Over time, workers should be able to find a better quality job, either at their current employer or by switching firms.

The study of the variation in hours around involuntary part-time workers concludes this section on how constrained workers sort across firms and jobs. A key takeaway is the non-negligible co-existence of part-time and full-time workers within the same jobs, which implies that hours are not solely determined by firm-level or occupation-level requirements. Worker-level characteristics also play a role in shaping hours allocation, a pattern that motivates the theoretical framework developed in Section ??.

Table 1: Within-Firm Determinants of Hours Constraints

Dependent Variable:	Age					Female	Tenure	Foreigner	Distance	Permanent
	15-24 (1)	25-34 (2)	35-44 (3)	45-54 (4)	55+ (5)	(6)	(7)	(8)	(9)	(10)
Involuntary Part-Time	0.025*** (0.004)	0.013* (0.005)	0.024*** (0.005)	-0.028*** (0.005)	-0.031*** (0.003)	0.143*** (0.005)	-0.179*** (0.016)	0.017*** (0.003)	-5.218*** (0.802)	-0.013*** (0.004)
Observations	699,991	699,991	699,991	699,991	699,991	699,991	699,991	699,991	699,991	699,991
Clusters	9,814	9,814	9,814	9,814	9,814	9,814	9,814	9,814	9,814	9,814
Adj. R ²	0.19	0.11	0.04	0.07	0.08	0.30	0.37	0.28	0.41	0.77
<i>Controls and Fixed Effects</i>										
Occupation × Workplace × Year	X	X	X	X	X	X	X	X	X	X
Age (Polynomial)						X	X	X	X	X
Female	X	X	X	X	X		X	X	X	X
Tenure	X	X	X	X	X	X		X	X	X
Foreigner	X	X	X	X	X	X	X		X	X
Distance	X	X	X	X	X	X	X	X		X
Permanent	X	X	X	X	X					

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05.

Note: This table presents differences in individual characteristics between involuntary part-time workers and their full-time coworkers. The data used is the EEC-DADS sample between 2003 and 2023. The sample is restricted to occupation × workplace combinations with at least one involuntary part-time worker. The sample of coworkers is restricted to full-time employees. Occupations correspond to the 2-digit level in the PCS classification (28 values). Occupations eligible to forfait jours contracts are removed from the sample. Age is split in 5 categories to account for non-linearities. Tenure corresponds to the number of years employed in current firm and is winsorized at 5 years due to data limitations. Foreigner is a dummy variable equal to 1 if the individual is born abroad. Distance corresponds to the distance to commute (in kilometers). Permanent is a dummy variable equal to 1 if the individual holds a permanent contract (*CDI*). Standard errors are clustered at the year × workplace × occupation level and reported in parentheses.

5 Labor Market Trajectories of Constrained Workers

This section provides first evidence on the evolution of constrained workers in the labor market. The depiction of constraints as mostly driven by occupation and firm effects suggests that workers would have to switch firms to align with their hour preferences. A central analytical focus is to understand to what extent constrained workers can improve their situation through mobility. Section 5.1 studies the relation of hours constraints to worker mobility and presents descriptive evidence on the evolution of hours and earnings. Section 5.2 introduces an event-study framework to measure differential changes in hours and earnings experienced by constrained workers. Section 5.3 explores the heterogeneity behind the main findings, both in terms of worker type and mobility channels.

The entire dynamic analysis relies on the EEC-DADS Panel introduced in Section 3.2.3 and represented in Figure A1. Similar restrictions to those used by Babet and Chabaud (2024) are applied to the sample³¹ and their definition of voluntary moves is used for robustness checks. Additionally, I restrict the sample to a balanced panel of observations between 3 years before and 3 years after the last year of the EEC period. After restrictions, the sample is composed of 157,716 workers. 16.2% report wanting to work more hours in the last quarter of their EEC period. This measure of hours constraints is used in the rest of the section and the year of the last EEC quarter serves as the reference year ($t = 0$). In short, the EEC-DADS Panel maintains the same structure as the DADS Panel, supplemented with additional information from the last year of EEC survey administration.

5.1 Worker Mobility

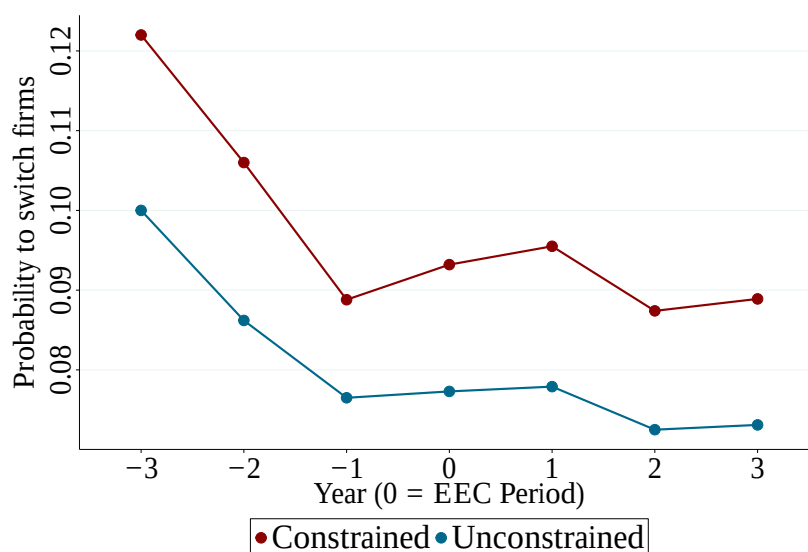
First, the focus is on the relation of hours constraints to mobility. The reference EEC employer is adjusted to the last employer reported in the EEC, thus counting any employer change right after the end of the EEC period as a move in $t = 1$. As discussed in Section 2, the relationship between hours constraints and mobility is theoretically ambiguous. On the one hand, constrained workers suffer a disutility from working short hours, which makes them more willing to move to a job offering more hours. On the other hand, because they are more employed in low-quality firms, their job arrival rate may be lower than that of unconstrained workers, which limits their ability to move to a better job. The empirical results below can therefore be read as informative about

³¹Workers aged between 18 and 64, stable working contract (*CDD* or *CDI*), annual hours above 120, hourly wage between 0.8 and 1,000 times the minimum wage, and focus on primary employment (defined as the spell that induces highest earnings).

which of these two forces dominates.

Figure 6 plots the probability of switching firms for constrained versus unconstrained workers, year by year around the EEC reference period. First, the group of constrained workers show a higher switching probability throughout the entire window, including the pre-period. Since this gap predates the EEC report, it likely reflects differences in characteristics between the two groups rather than a correlation with hours preferences. Yet, the gap widens around the EEC period: constrained workers' mobility reincreases after $t = 0$, while the share of moving unconstrained workers is stable through the post-EEC period. The group of constrained workers thus appear more likely to leave their firm specifically at the timing of the report, which is consistent with a disutility associated with being constrained in a short-hour jobs.

Figure 6: Worker Mobility in the EEC-DADS Panel



Note: This figure represents the probability of move to another firm by group of hours preferences during the EEC period. The x-axis shows years relative to the EEC period, while the y-axis displays the share of workers whose main employer is different from the year before. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period.

Table 2 shows the relation with controls on demographics (including occupation) and an employer fixed effect, thus abstracting from compositional differences between the groups. Mobility here is defined as a dummy equal to 1 if the worker switches employers at least once between $t = 1$ and $t = 3$ after the EEC period, a definition used for the rest of the paper. For this table only, the sample is also restricted to firms with at least 2 workers during the EEC period, so that an EEC employer fixed effect can be included.

The first column confirms that constrained workers are more likely to switch employers over the following three years by 2.3 pp, against a baseline mobility rate of 16%. Adding demographic and occupation controls cuts this effect roughly in half, and it disappears entirely once an employer fixed effect is added in the last specification. In other words, constrained workers do move more than the average worker, but not more than similar workers in the same firm, which fits with the compositional story suggested by the permanent gap in Figure 6. This last specification may, however, be too demanding given the data, and should be put into perspective: workers from the same firm are likely to face similar mobility frictions, regardless of whether they are constrained or not.

Table 2: Hours Constraints and Worker Mobility

Dependent Variables:	<i>Move in Years 1-3</i>		
	(1)	(2)	(3)
Constrained	0.023*** (0.005)	0.014** (0.005)	0.008 (0.005)
<i>Controls and Fixed-effects</i>			
EEC Year	X	X	X
Part-/Full-Time (EEC Period)	X	X	X
Demographics (EEC Period)		X	X
Employer (EEC Period)			X
Baseline	0.16	0.16	0.16
Share of Constrained	0.15	0.15	0.15
Observations	52,860	52,860	52,860
Adj. R ²	0.00	0.03	0.38

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05.

Note: This table presents estimates of the relationship between groups of hours preferences and job mobility. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The sample is also restricted to firms with at least 2 workers. The dependent variable is a dummy variable equal to 1 if the individual switches employers between 1 and 3 years after the EEC period. Demographics include age, gender, occupation and county of residence. Occupations correspond to the 2-digit level in the PCS classification (28 values). Standard errors are heteroskedasticity-robust and reported in parentheses.

The next inquiry is whether constrained workers who switch employers tend to increase their hours. Figure 7 illustrates the trajectories of weekly paid hours worked for four distinct groups constrained workers and unconstrained workers who switch employers at least once between periods 1 and 3 (respectively *constrained-movers* and *unconstrained-movers*), and their equivalent who remain with the same employer (*constrained-stayers* and *unconstrained-stayers*)³². The figure separates workers who were part-time (a) or full-time (b) during their EEC period.

³²The sample divides across the four groups as follows: 5.2%, 16.0%, 14.3%, and 64.5%.

In Figure 7a, constrained workers who switch employers experience a large increase in hours through mobility, closing most of the gap with the other groups by $t = 2$ and $t = 3$. By contrast, constrained workers who stay in the same firm see their hours increase modestly after their EEC period. Both types of unconstrained workers, whether movers or stayers, work more hours at the beginning of the period but their hours rise less than those of their equivalent constrained group. Despite the upward evolution for constrained part-time workers, their average hours remain relatively low at $t = 3$, while the majority of them reported wanting to secure a full-time job. Figure 7b does not show the same picture: all four groups follow a broadly similar downward trend in hours after $t = -1$, with constrained and unconstrained workers, movers and stayers alike, converging towards a similar level of hours by the end of the window.

For part-time workers, results support the argument that hours constraints drive job mobility as constrained workers who switch firms appear to increase their hours towards their desired level, while those who remain in the same firm show more stability. This contrast suggests that part-time workers classified as constrained cannot adjust their hours within their current firm, and largely rely on switching employers to do so. For full-time workers, the absence of a clear differential pattern suggests that hours constraints play a smaller role in shaping job mobility, likely because their hours preferences are less stringent, as previously emphasized in Section 4. Another potential explanation is that full-time workers cannot easily increase their hours as overtime hours are rarely advertised by employers.

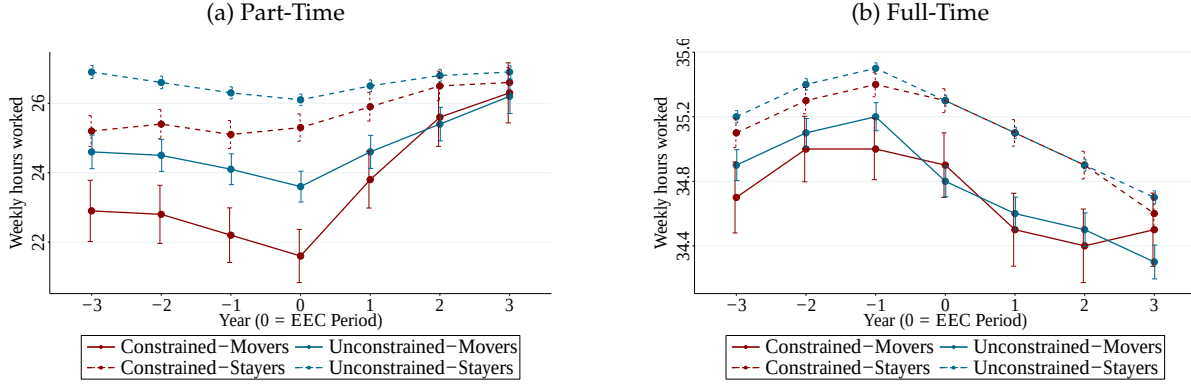
5.2 Event-Study Design

This section studies the evolution of constrained workers as compared to unconstrained workers using a more formalized approach. Given that constrained workers are mostly able to adjust their hours by switching firms, I estimate the differential change in outcome associated with employer-to-employer mobility for both types of workers. Hence, I compare the evolution of hours of constrained and unconstrained workers, conditional on moving. I use an event-study design summarized by the following equation:

$$y_{it} = \beta_1 \times S_{it} + \beta_2 \times S_{it} \times C_i + \alpha_i + \gamma_t + \varepsilon_{it} \quad (10)$$

The considered outcomes y_{it} are hours worked, hourly wages, and earnings (in logs). S_{it} is a

Figure 7: Evolution of Hours in the EEC-DADS Panel



Note: The figures represent the evolution of weekly hours across employer-to-employer moves for different worker groups, separately for workers who work part-time or full-time in the last quarter of the EEC period. The x-axis shows years relative to the EEC period, while the y-axis displays Hours are paid hours recomputed as full-year equivalent and divided by 52. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. Four distinct groups are tracked: constrained-movers (solid red line), unconstrained-movers (solid blue line), constrained-stayers (dashed red line), and unconstrained-stayers (dashed blue line). Workers are classified as constrained or unconstrained based on their reported hour preferences in the last quarter of the EEC period, and as movers or stayers based on whether they switch employers during years 1 to 3 after the EEC period. Error bars correspond to 95% heteroskedasticity-robust confidence intervals.

dummy variable equal to 1 if the individual i is a "mover" in year t , where t must be between 1 and 3 years after the EEC period. This restriction on considered moves favors their consistency with hours preferences reported in the EEC³³. C_i is a dummy variable equal to 1 if the individual i is constrained in the last quarter of their EEC period. α_i and γ_t are respectively worker-specific and year-specific fixed effects. β_1 captures the effect on the outcome associated with employer-to-employer moves. β_2 measures the differential effect for constrained workers. Worker fixed effects are included to control for time invariant unobserved heterogeneity.

Event-study designs are a widely used method for estimating treatment effects. Their validity in capturing causal effects yet relies on two critical identifying assumptions: no-anticipation and parallel trends (Borusyak et al., 2024). The no-anticipation assumption requires that workers do not change their behavior before switching jobs, while the parallel trends assumption holds if workers who change employers or stay would have followed similar trajectories in outcome absent mobility. To test the validity of the assumptions, I examine the similarities of movers and stayers groups in trend during the pre-event window. Figure A5a plots event-study coefficients for hours worked around employer moves and provides direct evidence of pre-trend violations. The pre-event coefficients, though modest in magnitude, are statistically significant and exhibit a notable

³³In addition, in the case where the individual experiences multiple transitions shortly after the EEC period, I only consider the first move as it is the one consistent with reported hours preferences.

decline in hours just prior to the move. This pre-move drop suggests that workers experiencing a relative reduction in hours are more likely to switch employers, consistent with an [Ashenfelter \(1978\)](#)'s dip argument. Figure [A5b](#) shows the equivalent coefficients for the interaction term by comparing constrained and unconstrained workers who switch firms. Both groups exhibit similar pre-trends in hours, suggesting that being constrained is not associated with different prior evolution of hours conditional on being a mover. More generally, the parallel trends assumption is not credible here as both employer-to-employer moves and worker preferences are likely endogenous to prior dynamics in working hours and earnings. Then, estimates from this specification cannot be interpreted as treatment effects of mobility. I use the event-study framework as a descriptive tool, as it provides a clear and intuitive quantification of how working hours of constrained and unconstrained workers diverge around employment transitions.

Table [3](#) reports estimates from the event-study model in Equation [10](#), estimated separately for workers who were part-time or full-time during their EEC period. The first row reports the effects of employer-to-employer moves for unconstrained workers. Among part-time workers, switching firms is associated with a 2.4% increase in hours, driven by a 13 pp higher probability of moving to a full-time job, while no such effect appears for full-time workers. Unconstrained movers yet experience a relative decrease in wages, which implies a significant relative reduction in earnings for full-time movers. This reduction may appear as surprising, but (a) it must be interpreted as relative to unconstrained stayers whose trajectories in earnings are increasing over time, (b) this result is consistent with [Babet and Chabaud \(2024\)](#), who study wage dynamics around job switches using the same data and find that cuts in annualized earnings are common even among voluntary moves, and likely rationalized by changes in non-wage amenities³⁴.

The second row reports the differential effect of mobility for constrained workers, the main focus of this paper. For part-time workers, in line with evidence in Figure [7](#), constrained movers experience larger increases in hours and earnings than unconstrained movers. Their hours increase by 5.9 pp more, reaching a total gain of 8.3% relative to non-movers, driven by a 17 pp higher probability of moving to a full-time job. Earnings of constrained movers also increase relatively more, by 5.4 pp, which implies a 3.8% increase as compared to non-movers, despite a reduction in wages similar than the one experienced by unconstrained movers. Their larger rise in hours thus

³⁴Voluntary moves are defined as moves out of a permanent contract (*CDI*) that did not involve a significant unemployment spell. This definition is used as a robustness check at the end of the section and shifts estimates upward for unconstrained movers.

Table 3: Event-Study Effects on Hours, Wages and Earnings

Group (EEC Period): Dependent Variable (in logs):	Part-Time			Full-Time		
	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Mover	0.024*** (0.009)	-0.039*** (0.004)	-0.016 (0.009)	-0.002 (0.002)	-0.046*** (0.001)	-0.048*** (0.002)
Mover $\times$ Constrained	0.059** (0.019)	-0.005 (0.007)	0.054** (0.018)	0.004 (0.005)	0.004 (0.004)	0.008 (0.005)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Baseline (in levels)	25.4	13.8	18,341	34.6	16.5	29,595
Observations	60,413	60,413	60,413	483,266	483,266	483,266
Adj. R ²	0.79	0.91	0.89	0.56	0.93	0.91

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The first three columns of the table correspond to estimates when restricting the sample to workers who primarily worked as part-time during the last year of their EEC period. The last three columns correspond to results in the equivalent full-time sample. The dependent variables are respectively the logs of weekly hours, hourly wages and net annual labor earnings from the DADS data. Hours are paid hours recomputed as full-year equivalent and divided by 52. Earnings are also recomputed as full-year equivalent based on the number of days worked over the year. "Mover" is a dummy variable equal to 1 if the worker's primary employer is different than the year before. The primary employer is defined as the one with highest annual earnings. In addition, moves used for the estimation must occur between 1 and 3 years after the EEC period. "Constrained" is a dummy variable equal to 1 if the worker is constrained (from above) in their hours in the last year of their EEC period. Worker fixed effects (Worker FE) control for time-invariant unobserved heterogeneity. Year fixed effects (Year FE) control for common time trends. Standard errors are heteroskedasticity-robust and reported in parentheses.

offsets the relative drop in wages common to all movers. Hence, employer-to-employer mobility tends to be an effective way for constrained part-time workers to raise their earnings. Among full-time workers, constrained movers experience small non-significant gains in hours and earnings, so that, combined with the negative baseline effect of mobility for this group, their earnings remain 3.4% below those of non-movers.

As discussed earlier, part-time employment is the most stringent form of hours constraints in the French labor market, and it is precisely among these workers that mobility yields the largest relative gains in hours and earnings. The estimates imply an 8.3% increase in hours relative to non-movers, which amounts to about 2 additional hours per week on average. This pattern is consistent with the idea that constrained part-time workers are more likely to move jobs specifically in pursuit of additional hours and higher earnings, rather than for other reasons such as amenities, career progression, or match quality. In other words, their mobility is more purposeful from an hours perspective.

To end this section, I evaluate whether three potential measurement issues affect the validity of the results. Reassuringly, the key findings of the section are robust to alternative specifications.

Voluntary Moves. The interpretation of job mobility as reflecting a preference for more hours relies on the assumption that these moves are voluntary. To address this point, I re-estimate the main specification from Equation 10 using a stricter definition of mobility. A move is classified as voluntary if it originates from a permanent contract and involves less than 60 days between jobs³⁵. Table A4 reveals stronger effects than in the original specification. Voluntary moves are, in the baseline case, associated with a larger increase in hours and smaller wage reductions. As a result, the combined effect on earnings becomes positive for part-time workers and smaller in magnitude for full-time workers. For constrained part-time workers specifically, voluntary moves enable a 15% increase in hours and a 10% increase in earnings as compared to non-movers. Conversely, there is no differential change between constrained and unconstrained movers in the full-time sample. This check thus provides a cleaner picture of the ability for constrained part-time workers to increase their hours through job mobility.

Stable Hours Preferences. As discussed in Appendix B1, workers' hours preferences show limited persistence over time. This raises the possibility that hours constraints measured during the EEC period may not persist afterward, i.e. some constrained workers may, in fact, no longer wish to increase their hours after $t = 0$. To limit this concern, I restrict the sample of constrained workers to those who consistently report wanting to work more hours during their EEC period, thereby focusing on workers with more stable preferences. Table A5 shows estimates broadly similar to the original specification. First row estimates remain nearly unchanged as the sample of unconstrained workers is unaffected by the restriction. For constrained workers, the interaction coefficients on hours and earnings are attenuated for part-time workers but roughly doubled for full-time workers. It is plausible that the restriction on stable preferences changes the profile of constrained workers, in particular for full-time workers, for whom the persistence of their EEC report may be driven by stronger constraints.

Period-Specific Estimates. To address the concern that the estimates may be driven by specific time periods, I split the sample into four subsamples, each excluding a different five-year period,

³⁵This definition differs slightly from Babet and Chabaud (2024), as I use the starting and ending dates of employment rather than unemployment benefit receipts to determine the no-unemployment criterion.

and re-estimate the main results. Table A6 shows a remarkable consistency in estimates across panels for constrained workers. The main exception is a notable decrease in the hours and earnings coefficients (in the first row) for part-time workers when years after 2018 are excluded. It means that estimates are larger in the recent period, i.e. part-time workers have increasingly moved towards jobs with long hours in recent years. Crucially, the drop in the last panel concerns unconstrained movers, but the differential change between constrained and unconstrained movers remains significant

Distance to Work. To limit the role of non-wage amenities other than hours, I re-estimate the main specification restricting the sample to movers who increase their commuting distance by less than 10 kilometers (or reduce it). Distances between municipalities are calculated using the Haversine formula, which computes the great-circle distance between two points on a sphere given their latitude and longitude coordinates. As compared to the original specification, Table A7 displays slightly larger changes in hours and earnings for unconstrained movers. On the other hand, the coefficients of the interaction term are smaller so that the combined effect for constrained workers is similar in magnitude to the original estimates.

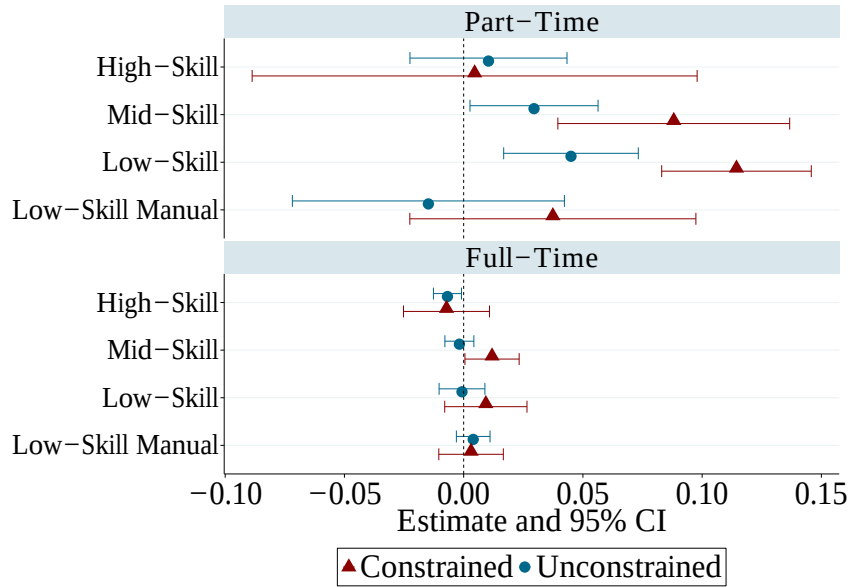
The main finding that constrained part-time workers gain significantly more in hours and earnings when they change jobs is robust across all checks. The next section explores the heterogeneity behind this result, focusing on different types of workers and job mobility.

5.3 Heterogeneity

The rest of this section explores heterogeneity in the effects across worker and mobility types, and asks whether constrained workers manage to close the gap to their desired hours over time.

Worker Type. Figure 8 breaks down the main estimates by broad occupation during the EEC period, using the first digit of the PCS classification. For constrained part-time workers, hours gains are primarily concentrated in low-skill and mid-skill non-manual occupations. Workers in low-skill non-manual jobs, who account for 55% of constrained part-time workers, experience an 11% increase in hours relative to non-movers, well above the 4.5% increase for their unconstrained counterparts, alongside a 7% relative increase in earnings. For full-time workers, as in Table 3, estimates are insignificant across all occupation groups.

Figure 8: Event-Study Effects by Type of Workers



Note: This figure shows event-study estimates on hours worked for constrained (red triangles) and unconstrained (blue circles) workers by type of worker. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The specification differs from Equation 10 (and Table 3) as the interaction term is excluded and separate regressions are run on the move dummy variable (S_{it} in Equation 10) for constrained and unconstrained workers. In each case, the group of non-movers is composed of both constrained and unconstrained workers. Occupations respectively correspond to the 1st-digit 3, 4, 5 and 6 in the PCS classification.

Mobility Type. Figure 9 turns to heterogeneity in job moves by examining two dimensions of employer-to-employer mobility: (i) whether it implies an occupational switch and (ii) the change in hours policies at the firm level, measured by AKM firm effects on hours. The first two rows split between-occupation movers by whether they move to an occupation with higher ($\uparrow h_{PCS}$) or lower ($\downarrow h_{PCS}$) hours fixed effects than their origin occupation. The last two rows split all movers by whether they move to a firm with a higher ($\uparrow h_{AKM}$) or lower ($\downarrow h_{AKM}$) AKM firm effect on hours than their origin firm.

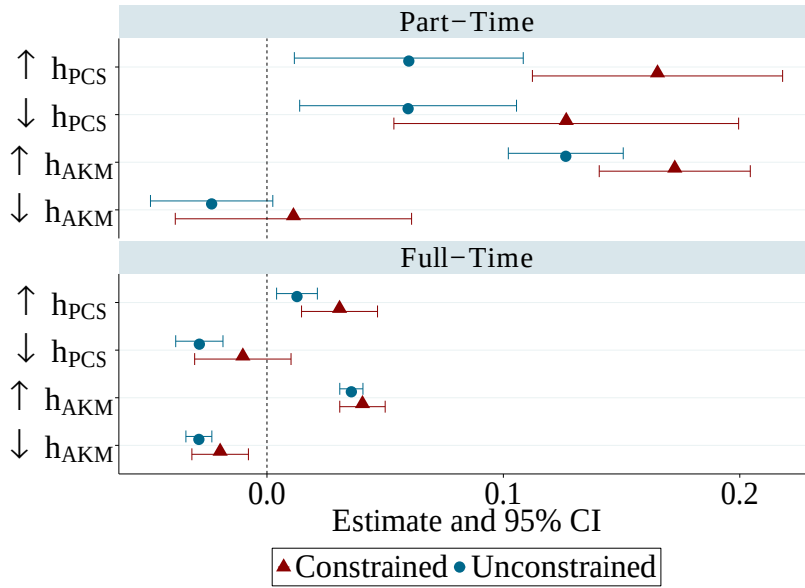
Starting with occupations, one natural interpretation of the large increase in hours for constrained part-time workers is that they adjust by moving to different occupations with longer hours requirements. This intuition directly follows from the strong concentration of hours constraints in a limited number of occupations shown in Section 4.1, consistent with previous research (Altonji and Paxson, 1986). For workers to increase their hours, they would need to switch occupations. The evidence from labor market trajectories does not entirely support this view. Table A8 in Appendix shows that occupation-to-occupation mobility is indeed associated with more hours gains

than within-occupation moves, particularly for constrained part-time workers (12% versus 6.5%). However, Figure 9 indicates that the direction of the switch, in terms of PCS hours requirement, barely matters: hours increase by more than 10% in both directions, and the coefficients are not statistically different. This result suggests that constrained part-time workers accept to move to a different occupation whenever they find a job with long hours, regardless of the typical hours requirements in the given occupation. Occupation-to-occupation moves are arguably more costly to a worker, e.g. learning costs, so they must generate larger gains for workers to undertake them. For full-time workers, for whom variation in hours is smaller in a given occupation, the direction of occupation-to-occupation mobility matters more. Moves to higher-FE occupations are associated with a significant 3% increase in hours, while moves to lower-FE occupations show no significant effect.

Turning to firm-level hours policies, the picture is strikingly clear. For all workers, moving to a higher h_{AKM} firm raises hours worked, while moving to a lower h_{AKM} firm produces either negative or negligible changes. Constrained and unconstrained part-time movers are distributed similarly across these two types of moves (60% versus 54% move to higher h_{AKM} firms), and their changes in hours worked are in the same order of magnitude. Constrained part-time workers who move to higher h_{AKM} firms experience a 17% increase in hours, which entirely drives the 8.3% estimate from Table 3. The destination firm's hours policy thus appears to be a key determinant of hours gains upon mobility. This is very much in line with the job ladder argument developed in Section 2: constrained workers increase their hours, then earnings, by climbing towards firms with longer hours policies, and job mobility is the main vehicle through which this reallocation takes place.

Hours Mismatch. Beyond job mobility, I ask to what extent constrained workers manage to close the gap to their self-reported desired hours through their move. I restrict the sample to constrained workers and track the share of them who reach their preferred hours within three years after the report. Results show that only 18% of constrained part-time workers and 4% of full-time workers achieve to do so. The lower rate for full-time workers reflects their unrealistic desired hours, given the limited variation in hours across full-time jobs. The proportion of workers reaching desired hours also declines through event times, which means that some workers are able to adjust hours in the short-run, as for them hours constraints are a transitory issue, while others cannot find jobs with long hours, thus remain constrained in the following years. This section temper the earlier

Figure 9: Event-Study Effects by Type of Mobility



Note: This figure shows event-study estimates on hours worked for constrained (red triangles) and unconstrained (blue circles) workers by type of mobility. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The specification differs from Equation 10 (and Table 3) as the interaction term is excluded and separate regressions are run for constrained and unconstrained workers to estimate β_1 , using the same group of non-movers. See Table 3 for details about the variable used for mobility. The sample of movers is alternatively restricted to specific types of mobility. The first two rows show estimates for between-occupation movers split by whether they move to an occupation with more ($\uparrow h_{PCS}$) or fewer ($\downarrow h_{PCS}$) hours than their origin occupation. The last two rows show estimates for all movers (both within-occupation and between-occupation) split by whether they move to a firm with a higher ($\uparrow h_{AKM}$) or lower ($\downarrow h_{AKM}$) AKM firm effect on hours than their origin firm. The estimation of AKM firm effects is detailed in Section 4.2. Non-movers serve as the comparison group for all estimates. Error bars correspond to 95% heteroskedasticity-robust confidence intervals.

estimates. While constrained workers tend to significantly increase their hours on average, a large majority does not manage to close the gap to their preferred hours level.

Figure A6 in Appendix examines what predicts the probability of reaching desired hours within three years, separately for part-time and full-time constrained workers. Age and being a woman both reduce this probability, particularly among part-time workers. Occupation matters differently across groups: mid-skill workers are more likely to reach their desired hours among part-time workers, while low-skill manual workers (the reference group) have better probabilities in the full-time group. As expected from previous results, employer-to-employer mobility is the most effective mechanism for increasing hours, regardless of whether the move involves an occupational switch. In addition, workers with smaller gaps between actual and desired hours are, naturally, more likely to close them. Overall, the workers least likely to escape hours constraints are also the most vulnerable: older workers, women, and those in low-skill part-time jobs, for whom underemployment is both harder to escape and likely to be most costly.

Stepping back, this section shows that constrained workers can increase their hours and earnings through certain types of job mobility, but it is far from a universal remedy. Most constrained workers do not move, and for those who do, a large majority never reaches their preferred hours level. This points to hours constraints as a persistent and structural feature of the labor market, one that job mobility can mitigate but not resolve. If workers cannot easily escape this situation, one may wonder how much this situation costs them. The next section turns to this question and estimates the welfare impacts associated with hours constraints.

6 Willingness-To-Pay

This section aims to identify workers' willingness-to-pay (WTP) to adjust working hours to their desired level, building on the approach of [Le Barbanchon et al. \(2020\)](#). The goal of the estimation is to quantify in monetary terms the welfare costs associated with hours constraints. I start by introducing the framework in Section 6.1 and then describe the estimation procedure as well as the results in Section 6.2.

6.1 Identification

The identification draws on a simple framework is embedded in the theoretical model introduced in Section 2. Following [Lachowska et al. \(2026\)](#), the utility of worker i is specified as $u_i(e, h) = e - \varepsilon_i h^\mu$, where ε_i captures preferences for leisure and μ measures disutility from working, with $\mu > 1$. By definition, earnings are equal to the product of wage and hours: $e \equiv wh$, so that utility can be rewritten: $u_i(w, h) = wh - \varepsilon_i h^\mu$. Suppose that workers are asked (as in the EEC) to report their desired level of hours h_i^* at their given wage level $w_{i,0}$. In this framework, worker i 's willingness-to-pay to work their preferred hours corresponds to the gap between the current wage and the wage that would provide the same utility level when working this level of hours:

$$WTP_i \equiv w_{i,0} - w_{i,C} \quad (11)$$

where $w_{i,C} = \frac{u_i(w_{i,0}, h_i, 0) + \varepsilon_i h_i^*}{h_i^*}$ represents this counterfactual wage at h_i^* with utility $u_{i,0}$.

The WTP parameter measures the absolute reduction in the (hourly) wage rate that a worker would accept in exchange for working their preferred hours. For underemployed workers, this wage reduction is more than offset by the implicit increase in hours worked, so that earnings from

the counterfactual bundle with preferred hours are higher than current earnings $e_{i,0} \equiv w_{i,0}h_{i,0}$. Expressing WTP in hourly wage terms is therefore subtle: the parameter captures a willingness to forego part of a potential earnings gain, not a willingness to lose in absolute earnings.

The following example illustrates this point. Suppose a worker currently works 20 hours per week for €250 per week (or €1,000 per month) and reports wanting to work 40 hours per week with a corresponding variation in earnings. At the current hourly wage, doubling hours worked would yield €500 per week (or €2,000 per month). Yet, it is plausible that this worker would accept to work 40 hours for *less* than €500. The question that the estimation aims to answer is how much *less*. If the worker's acceptance threshold is €450 per week, then the WTP to work 40 hours per week equals $(500-450)/40 = €1.25$ per hour, that is 10% of the baseline hourly wage. This amount reflects the monetary value of the utility gain from hours adjustment: the worker is willing to trade €1.25 per hour in order to move to their preferred hours.

The estimation challenge is to recover the utility parameters needed to identify WTP_i . A key simplification arises from the optimality of the bundle $(w_{i,0}, h_i^*)$: at this point, the marginal utility with respect to hours equals zero, which allows ε_i to be expressed as a closed-form function of μ , $w_{i,0}$, and h_i^* (Equation 12). The estimation therefore reduces to a single unknown parameter, μ .

$$\left. \frac{\partial u_i}{\partial h} \right|_{h=h_i^*, w=w_{i,0}} = 0 \Leftrightarrow \varepsilon_i = \frac{w_{i,0}}{\mu h_i^{*\mu-1}} \quad (12)$$

The parameter μ plays a central role in the WTP estimation, as it entirely governs the curvature of the disutility of work, i.e. is, how rapidly marginal disutility increases with hours. It is assumed common across workers and directly determines the elasticity of labor supply, which equals $1/(\mu - 1)$ under this specification³⁶. On the other hand, the parameter ε_i is worker-specific and controls the level of disutility associated with working more hours. This specification thus allows to disentangle a common response to additional hours of work from idiosyncratic heterogeneity in hours preferences across workers. In addition, as μ is common to all workers (and ε_i is derived from Equation 12), differences in the WTP parameter across workers are entirely driven by differences in labor market conditions $(w_{i,0}, h_{i,0})$ and preferred hours h_i^* .

To estimate μ , I exploit employer-to-employer moves and the wage–hour bundles $(w_{i,1}, h_{i,1})$ ob-

³⁶See Appendix C.2 for the proof.

served at the destination job, building on [Le Barbanchon et al. \(2020\)](#)’s approach to valuing commuting distance. Their method estimates the parameters of the iso-utility curve by minimizing the distance of downward-utility moves to the iso-utility curve. The underlying assumption is a revealed preference criterion: a worker accepts a new job only if it yields a higher utility than the current one. The estimated iso-utility curve is thus the one that best rationalizes observed job mobility, i.e. the one that implies $u(w_{i,1}, h_{i,1}) > u(w_{i,0}, h_{i,0})$ for most workers.

Figure 10 illustrates the estimation strategy in the hours-wage plane. The original job bundle $(w_{i,0}, h_{i,0})$ and the worker’s optimal hours h_i^* are shown. Green points represent destination bundles for workers sharing the original job bundle $(w_{i,0}, h_{i,0})$ and optimal hours h_i^* . The objective is to estimate μ , and ε_i , so as to recover the y -axis coordinate of the blue point, which yields the WTP parameter. The black curve represents the iso-utility curve through $(w_{i,0}, h_{i,0})$ that best fits the observed employer-to-employer moves. The red curve shows the iso-utility curve through the optimal bundle $(w_{i,0}, h_i^*)$. Note that in practice, the parameter μ is simultaneously estimated on the entire sample of workers, with no requirements over the number of destination bundles for a given set of values $(w_{i,0}, h_{i,0}, h_i^*)$.

The optimization problem directly parallels [Le Barbanchon et al. \(2020\)](#)’s estimation of the slope parameter in the wage–commute plane. For a given μ , let $\mathcal{B}_\mu$ denote the set of destination bundles that fall below the current job’s iso-utility curve: $\mathcal{B}_\mu = \{i \mid u(w_{i,1}, h_{i,1}) < u(w_{i,0}, h_{i,0})\}$. The estimator of μ minimizes the distance of such revealed-preference violations to the iso-utility curve:

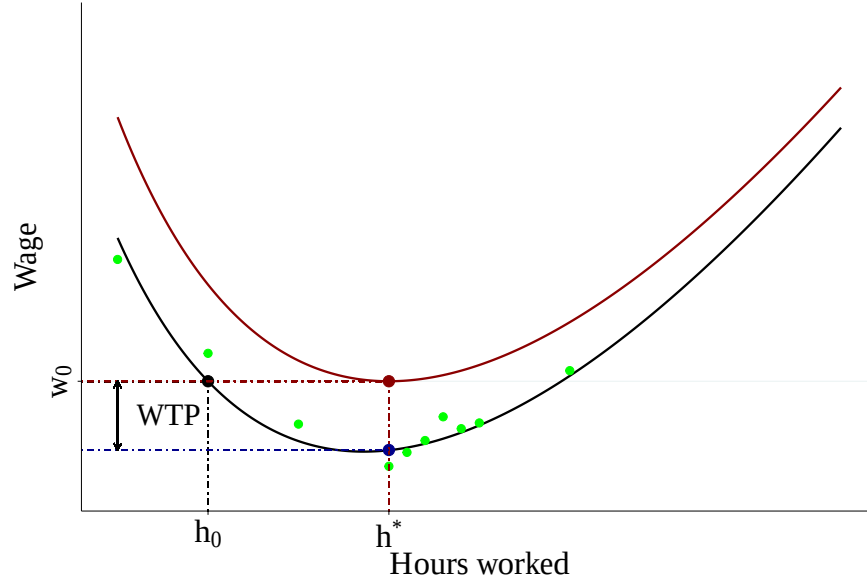
$$\hat{\mu} = \operatorname{argmin}_{\mu} \sum_{i \in \mathcal{B}_\mu} d_{\mu, w_{i,0}, h_{i,0}}(w_{i,1}, h_{i,1})$$

where $d_{\mu, w_{i,0}, h_{i,0}}(w_{i,1}, h_{i,1})$ denotes the Euclidean distance between the destination bundle $(w_{i,1}, h_{i,1})$ and its orthogonal projection onto the iso-utility curve of parameter μ passing through $(w_{i,0}, h_{i,0})$. In other terms, $d_{\mu, w_{i,0}, h_{i,0}}(w_{i,1}, h_{i,1})$ captures how far the move from $(w_{i,0}, h_{i,0})$ to $(w_{i,1}, h_{i,1})$ is from respecting the revealed preference criterion.

6.2 Estimation

This section describes the implementation of the estimation strategy. I restrict the EEC-DADS Panel to constrained workers who switch firms between 1 and 3 years after their EEC period, starting from a permanent contract. This yields a sample of 3,926 workers. As in previous sections,

Figure 10: Willingness-To-Pay Framework



Note: The figure illustrates the framework for estimating workers' willingness-to-pay (WTP) to reach their desired hours. The original job bundle $(w_{i,0}, h_{i,0})$ and the worker's optimal hours h_i^* are shown. Green dots represent observed destination bundles $(w_{i,1}, h_{i,1})$. The estimation method identifies utility parameters by minimizing the Euclidean distance between destination bundles below the iso-utility curve and their projections onto it. The black curve represents the iso-utility curve through $(w_{i,0}, h_{i,0})$ that best fits the observed employer-to-employer moves. The red curve shows the iso-utility curve through the optimal bundle $(w_{i,0}, h_i^*)$. The WTP parameter is the vertical distance between the wage at the blue point and $w_{i,0}$, representing the wage reduction a worker would accept to work optimal hours while maintaining utility.

in the case where a worker moves multiple times over this period, I restrict to the first move to ensure consistency with reported hours preferences. For external validity, Table 4 compares this sample of movers to the broader population of constrained workers in the EEC-DADS Panel. The two samples are broadly comparable across age, gender, occupation, and hours distributions, with the WTP estimation sample slightly over-representing prime-age workers (25–44) and under-representing the youngest and oldest age groups, which is expected given the requirement of an employer-to-employer transition.

Table 4: Representativity of Constrained Workers in the WTP Estimation Sample

Variable	WTP Sample	EEC-DADS Panel
N	3,926	26,496
Age - 15-24 (%)	2.3	5.4
Age - 25-34 (%)	37.7	30.5
Age - 35-44 (%)	31.8	30.2
Age - 45-54 (%)	22.4	25.7
Age - > 55 (%)	5.9	8.3
Female (%)	37.4	41.3
Occupation - High-Skill (%)	12.0	10.9
Occupation - Mid-Skill (%)	33.0	30.0
Occupation - Low-Skill Non-Manual (%)	26.3	27.9
Occupation - Low-Skill Manual (%)	28.7	31.2
Paid Hours Worked - < 25	8.3	11.6
Paid Hours Worked - 25-34	33.6	31.0
Paid Hours Worked - 35-39	50.8	52.3
Paid Hours Worked - > 40	6.9	5.0

Note: This table compares the WTP estimation sample of constrained workers who switch firms within three years of their EEC interview to the full sample of constrained workers in the EEC-DADS Panel. All variables are measured in the last year of the EEC period.

I solve the optimization problem by using a grid search procedure. The reason is that the objective function is not available in closed form as a function of μ , and cannot be directly optimized. For any candidate value of μ , computing the distance $d_{\mu, w_{i,0}, h_{i,0}}(w_{i,1}, h_{i,1})$ requires first solving a numerical inner problem which consists of projecting each destination bundle onto the iso-utility curve induced by that value of μ . The set of violations $\mathcal{B}_\mu$ also depends on μ , as the classification of a destination bundle in the downward-utility group is itself determined by the candidate iso-utility curve. As a result, the mapping from μ to the objective function value cannot be derived analytically. The estimator is therefore computed by evaluating the objective function over a fine grid of candidate values of μ and selecting the value that minimizes the total objective function over all workers.

For a given value of μ , the procedure proceeds as follows:

- (i) For each worker i , characterized by their original bundle $(w_{i,0}, h_{i,0})$ and preferred hours h_i^* , I derive ε_i from Equation 12. Given μ and ε_i , the iso-utility curve through $(w_{i,0}, h_{i,0})$ is fully determined.
- (ii) I retain all employer-to-employer moves for which the destination bundle yields a utility level strictly below $u_i(w_{i,0}, h_{i,0})$, thereby constructing $\mathcal{B}_\mu$.

(iii) For each destination bundle in $\mathcal{B}_\mu$, I compute the Euclidean distance to its orthogonal projection onto the iso-utility curve³⁷.

(iv) The objective function is the sum of distances across all workers in $\mathcal{B}_\mu$.

Steps (i) to (iv) are repeated for each value of μ on a grid from 1.01 to 10 in increments of 0.01, and $\hat{\mu}$ is taken as the minimizer. Once $\hat{\mu}$ is identified, each worker's WTP is recovered via Equations 11 and 12. To obtain standard errors, the entire procedure is bootstrapped by resampling workers with replacement and re-running the grid search on each bootstrap sample. This approach correctly propagates sampling uncertainty on considered employer-to-employer moves, yielding standard errors for $\hat{\mu}$ and, by extension, for the distribution of WTP estimates. A notable feature of this approach is that it requires no assumption, nor arbitrary parameter calibration, beyond the functional form of preferences.

The grid search estimation yields $\hat{\mu} = 5.76$. This corresponds to a Marshallian labor supply elasticity equal to 0.21, much in line with previous estimates (Kleven et al., 2025). Figure 11 shows the distribution of WTP estimates for all workers, relative to their original wage $w_{i,0}$. Estimates are particularly concentrated between 0 and 10%, and the average (represented by the solid red line) is equal to 10.2%. The dashed red lines show the 95% confidence interval around the mean, obtained by bootstrapping the full estimation procedure over 500 replications. The WTP is higher for part-time, about 15%, than for full-time workers, 8%, since part-timers have larger gaps to desired hours.

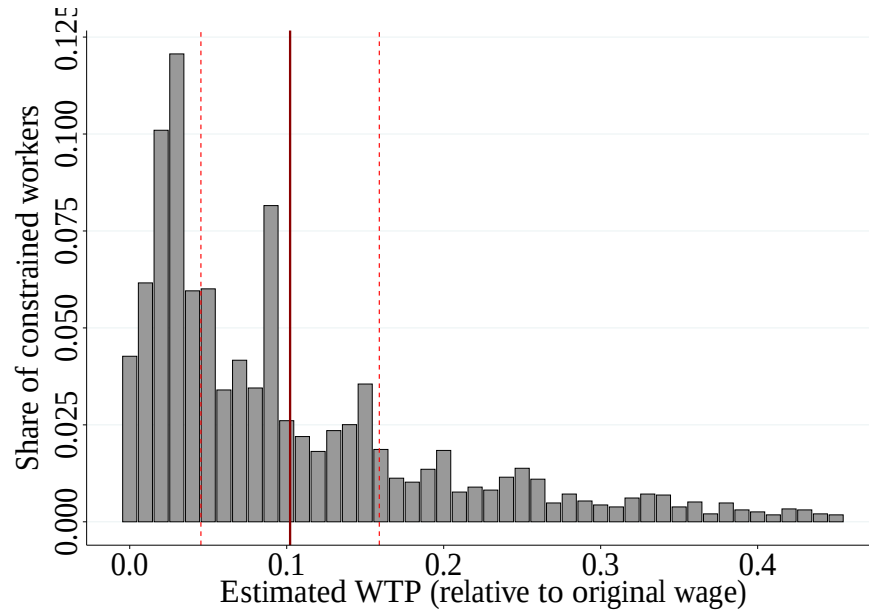
The estimated WTP parameter of approximately 10% of the baseline hourly wage is broadly consistent with the findings of Lachowska et al. (2026), who estimate that workers would require a 12% wage premium from their current employer to be as well off as they would be at an employer offering their ideal hours. The two definitions are nevertheless conceptually distinct³⁸. The WTP definition adopted here asks how much the hourly wage could be reduced if hours simultaneously moved to h_i^* , at a given utility level. Lachowska et al. (2026) instead ask how much the wage would need to change, at the worker's current hours, to increase utility to the optimal bundle level. Their approach thus does not identify a WTP parameter strictly speaking, but it provides a utility value of closing the hours gap.

An advantage of their approach is that it avoids a limitation inherent to mine: since h_i^* is reported as desired hours at the current wage $w_{i,0}$, the counterfactual bundle $(w_{i,C}, h_i^*)$ is likely not on the

³⁷The projection is obtained by solving for the point on the iso-utility curve at which the derivative of the squared distance with respect to hours equals zero.

³⁸See Appendix C.2 for a comparison of both definitions.

Figure 11: Willingness-To-Pay Estimates



Note: The figure shows the distribution of individual WTP estimates, expressed as a share of the baseline hourly wage $w_{i,0}$, for the 3,926 constrained workers in the WTP estimation sample. The solid red line marks the mean WTP (10.2%), with the dashed red lines showing the associated 95% confidence interval, obtained by bootstrapping the full grid search procedure over 500 replications.

worker's labor supply curve. Using the definition of [Lachowska et al. \(2026\)](#), I find that workers would require a 16% wage increase to obtain a utility gain equivalent to working their desired hours, a figure slightly higher than their estimate of 12%.

7 Conclusion

This paper leverages a unique data linkage to produce novel empirical evidence on hours constraints in France. In the French labor market, where the most salient frontier in working hours runs between part-time and full-time employment, involuntary part-time work stands out as the most prominent and policy-relevant form of hours constraints, as it traps workers in precarious employment.

By combining self-reported hour preferences from the Labour Force Survey with administrative employer-employee records, I am able to characterize the labor market context of constrained workers with greater precision than prior research. For instance, I can observe hours worked by comparable workers in the same firm, workplace, or occupation, and assess whether classic explanations, such as firm rigidities, account for underemployment. While firm sorting is non-

negligible, pure technological rigidities are unlikely to be the primary driver, as full-time work remains common in firms with involuntary part-time workers. This evidence also implies that variation in hours within a firm is not a good proxy for hours constraints.

The paper's main contribution is to provide the first evidence on workers' ability to close the hours gap through employer-to-employer mobility. On average, constrained workers achieve larger gains in hours and earnings when they change employers than both unconstrained movers and constrained non-movers. Yet, a large majority remain unable to move to a better job in the short run.

Given the persistence of hours constraints, in the final section, I use a revealed preference framework to estimate the welfare costs for workers from being off their labor supply curve. I find that constrained workers would accept, on average, a 10% reduction in their hourly wage to reach their preferred hours. This result is in the same order of magnitude as estimates by [Lachowska et al. \(2026\)](#) in the US, and reflects the high value of additional earnings as compared to work disutility for those workers.

Overall, this paper highlights underemployment as a multi-dimensional problem in modern labor markets that, like unemployment, stems from the interaction of worker-specific characteristics and firm-level factors. While workers' hours preferences are well understood, more evidence on firms' preferences over hours worked is needed to design effective policies to reduce underemployment and improve labor market matching.

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Appendix A Additional Figures and Tables

A.1 Tables

Table A1: EEC and Matched EEC-DADS Samples, 2003-2023

		EEC Sample	EEC Sample (with <i>SIRET</i>)	EEC-DADS Sample	EEC-DADS Panel
Number of observations		3,201,635	2,619,074	1,695,359	9,734,018
Number of workers		791,701	654,632	435,472 (67%)	399,553 (61%)
Number of firms		262,823	262,823	196,484	612,603
Full-Time (%)		81.0	82.7	82.8	82.8
Female (%)		50.4	49.3	48.4	49.4
Age	15-24	8.3	8.1	8.5	13.4
	25-34	20.6	20.8	21.3	24.6
	35-44	26.0	26.3	26.9	25.1
	45-54	28.3	28.4	27.9	24.1
	55+	16.7	16.3	15.5	12.8
Occupation (%)	High-Skill	17.4	18.1	17.3	17.4
	Mid-Skill	26.4	27.5	27.3	26.4
	Low-Skill Non-Manual	32.2	30.2	28.8	30.4
	Low-Skill Manual	24.0	25.0	26.6	25.7
Firm Size (%)	< 10	59.1	57.6	58.2	58.7
	10-49	15.4	16.0	16.6	17.3
	50-499	17.3	18.1	18.2	17.5
	≥ 500	8.2	8.3	7.0	6.5
Usual Hours Worked (%)	< 25	10.4	9.0	8.6	11.8
	25-34	10.0	9.9	9.7	10.0
	35-39	52.5	55.1	55.9	53.0
	40-49	19.5	18.9	18.9	18.0
	≥ 50	7.6	7.1	7.0	7.2
Share of constrained from above - Full-Time (%)		16.3	16.9	19.0	19.5
Share of constrained from above - Part-Time (%)		34.7	33.8	26.0	23.8
Share of involuntary part-time workers (%)		24.2	24.3	19.0	17.1
Share of constrained from below (%)		2.4	2.5	2.5	2.5

Note: This table compares the two newly built EEC-DADS and EEC-DADS Panel samples with the original EEC sample. The EEC sample (with *SIRET*) corresponds to the EEC sample after removing observations where the establishment's ID is missing. Shares of constrained and involuntary part-time workers are computed in proportion of the corresponding working time status considering the workers' last quarter of observation.

Table A2: Summary Statistics of Constrained Workers in EEC-DADS and EEC, 2003-2023

Variable	Involuntary Part-Time		Full-Time	
	EEC	EEC-DADS	EEC	EEC-DADS
Age	37.1	38.1	37.8	37.5
Female (%)	73.8	71.8	61.5	58.6
Occupation - High-Skill (%)	4.0	4.6	10.8	9.9
Occupation - Mid-Skill (%)	16.4	12.7	27.8	22.1
Occupation - Low-Skill Non-Manual (%)	58.3	60.0	27.4	31.1
Occupation - Low-Skill Manual (%)	21.4	22.7	33.9	36.9
Born Abroad (%)	13.8	12.7	10.8	10.5
Public Sector (%)	23.7	19.6	17.6	13.0
Permanent Contract (%)	58.4	71.7	83.3	83.2
Usual Hours Worked	23.3	26.4	37.3	35.6

Note: This table presents summary statistics of constrained workers in the EEC and EEC-DADS samples. "Occupation" categories correspond to the 1st-digit in the PCS classification, respectively 3, 4, 5 and 6. "Born Abroad" indicates the proportion of individuals born outside of France. "Public Sector" shows the proportion of workers in the public sector. "Permanent Contract" represents the share of workers with a permanent contract (*contrat à durée indéterminée*). Hours worked correspond to usual hours worked from the Labour Force Survey.

Table A3: Summary Statistics by Group of Workers' Preferences, 2003-2023

Variable	Constrained from above	Involuntary part-time	Constrained from below	Unconstrained
Age - 15-24 (%)	10.9	14.8	1.8	5.5
Age - 25-34 (%)	28.4	23.6	19.2	20.2
Age - 35-44 (%)	28.3	25.9	28.9	27.2
Age - 45-54 (%)	24.3	25.7	32.5	30.4
Age - > 55 (%)	8.2	10.0	17.6	16.7
Female (%)	48.7	77.2	61.2	49.1
Occupation - High-Skill (%)	8.9	4.6	31.4	20.2
Occupation - Mid-Skill (%)	24.9	16.0	32.2	28.3
Occupation - Low-Skill Non-Manual (%)	37.1	60.9	24.6	28.8
Occupation - Low-Skill Manual (%)	29.1	18.5	11.8	22.6
Workplace Size - < 10 (%)	21.6	31.8	15.0	17.4
Workplace Size - 10-49 (%)	31.0	35.3	26.4	28.5
Workplace Size - 50-499 (%)	33.2	25.5	36.7	35.4
Workplace Size - $\geq$ 500 (%)	14.3	7.3	22.0	18.7
Born Abroad (%)	11.9	14.9	8.2	10.3
Public Sector (%)	20.1	22.8	24.1	23.7
Permanent Contract (%)	80.9	65.8	94.9	91.8
Temporary Worker (%)	3.7	2.1	0.7	1.5
Secondary Job (%)	3.0	5.9	2.3	2.1
Share of Part-Time (%)	28.0	-	11.2	14.5
Paid Hours Worked (Full-Time)	36.1	-	36.2	36.2

Note: This table presents summary statistics by group of hours preferences in the EEC sample. Groups are considered based on their answers to the following questions in the surveys: "do you want to work [more/less] hours in your job with a corresponding income variation?". "Constrained from above" workers report that they want to work more hours ; "Involuntary part-time" workers are the subsample of the previous group who currently work part-time but would ideally work full-time based on their desired hours ; "Constrained from below" workers report that they want to work less hours ; "Unconstrained" workers answer no to both questions. "Occupation" categories correspond to the 1st-digit in the PCS classification, respectively 3, 4, 5 and 6. "Workplace Size" groups correspond to the number of salaried workers employed in the workplace. "Born Abroad" indicates the proportion of individuals born outside of France. "Public Sector" shows the proportion of workers in the public sector. "Permanent Contract" represents the share of workers with a permanent contract (*contrat à durée indéterminée*). "Temporary Worker" represents the share of workers with a temporary contract (*contrat d'intérim*). "Secondary Job" shows the proportion of workers with a secondary job. Restricting to the EEC-DADS sample, the final row reports average paid hours worked of workers with full-time contracts.

Table A4: Event-Study Effects on Hours, Wages and Earnings - Voluntary Moves

Group (EEC Period):	Part-Time			Full-Time		
Dependent Variable (in logs):	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Mover	0.059*** (0.012)	-0.023** (0.007)	0.037*** (0.012)	0.015*** (0.002)	-0.023*** (0.002)	-0.008** (0.003)
Mover $\times$ Constrained	0.094*** (0.026)	-0.030* (0.013)	0.064** (0.025)	0.005 (0.006)	-0.003 (0.004)	0.001 (0.006)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Baseline (in levels)	25.4	13.8	18,341	34.6	16.5	29,595
Observations	57,439	57,439	57,439	466,261	466,261	466,261
Adj. R ²	0.80	0.90	0.89	0.61	0.93	0.91

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10 using the alternative definition of employer-to-employer moves similar to [Babet and Chabaud \(2024\)](#). The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The sample is restricted to moves from a permanent contract with an unemployment period of less than 60 days between jobs. See Table 3 for details.

Table A5: Event-Study Effects on Hours, Wages and Earnings - Stable Constrained Workers

Group (EEC Period):	Part-Time			Full-Time		
Dependent Variable (in logs):	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Mover	0.024** (0.009)	-0.040*** (0.004)	-0.016 (0.009)	-0.002 (0.002)	-0.046*** (0.001)	-0.048*** (0.002)
Mover $\times$ Constrained	0.043** (0.017)	-0.005 (0.008)	0.039* (0.025)	0.016*** (0.005)	0.005 (0.004)	0.021*** (0.005)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Baseline (in levels)	25.4	13.9	18,438	34.6	16.6	29,782
Observations	56,125	56,125	56,125	455,484	455,484	455,484
Adj. R ²	0.78	0.89	0.88	0.60	0.93	0.90

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10 using an alternative definition of constrained workers with stable preferences over their EEC period. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. The sample of constrained workers is restricted to those whose preferences remain identical throughout their EEC period. See Table 3 for details.

Table A6: Event-Study Effects on Hours, Wages and Earnings - Specific Periods

Group (EEC Period): Dependent Variable (in logs):	Part-Time			Full-Time		
	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Panel A: 2009-2023						
Mover	0.025** (0.009)	-0.039*** (0.004)	-0.014 (0.009)	-0.001 (0.002)	-0.046*** (0.001)	-0.047*** (0.002)
Mover $\times$ Constrained	0.065*** (0.015)	-0.008 (0.007)	0.057*** (0.015)	0.007 (0.004)	0.004 (0.003)	0.011* (0.005)
Panel B: 2003-2008, 2014-2023						
Mover	0.035*** (0.010)	-0.046*** (0.005)	-0.011 (0.010)	0.003 (0.002)	-0.048*** (0.002)	-0.045*** (0.002)
Mover $\times$ Constrained	0.056** (0.017)	-0.002 (0.008)	0.054** (0.017)	0.003 (0.005)	0.003 (0.004)	0.006 (0.005)
Panel C: 2003-2013, 2019-2023						
Mover	0.040*** (0.012)	-0.045*** (0.006)	-0.005 (0.012)	0.003 (0.002)	-0.048*** (0.002)	-0.045*** (0.003)
Mover $\times$ Constrained	0.064** (0.021)	-0.002 (0.008)	0.062** (0.020)	0.009 (0.006)	0.003 (0.004)	0.012* (0.006)
Panel D: 2003-2018						
Mover	0.002 (0.012)	-0.030*** (0.006)	-0.028* (0.012)	-0.015*** (0.002)	-0.041*** (0.002)	-0.056*** (0.003)
Mover $\times$ Constrained	0.066** (0.021)	-0.020* (0.009)	0.046* (0.021)	0.011 (0.006)	0.005 (0.004)	0.016** (0.006)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Observations - Part-Time: A = 59,102 ; B = 46,769; C = 37,665; D = 37,706.						
Observations - Full-Time: A = 469,357 ; B = 356,885 ; C = 310,121; D = 313,447.						

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10 using using four subsamples, each excluding a different five-year period. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. See Table 3 for details.

Table A7: Event-Study Effects on Hours, Wages and Earnings - Distance to Work

Group (EEC Period): Dependent Variable (in logs):	Part-Time			Full-Time		
	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Mover	0.045*** (0.010)	-0.039*** (0.005)	0.006 (0.010)	-0.001 (0.002)	-0.041*** (0.002)	-0.042*** (0.003)
Mover $\times$ Constrained	0.047** (0.018)	-0.009 (0.009)	0.038* (0.018)	0.005 (0.005)	0.002 (0.004)	0.007 (0.005)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Baseline (in levels)	25.5	13.8	18,424	34.6	16.5	29,555
Observations	56,684	56,684	56,684	457,988	457,988	457,988
Adj. R ²	0.78	0.89	0.88	0.60	0.93	0.90

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10 after restricting the sample to moves that increase the distance to commute by less than 10 kilometers (or reduce it). The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. Distances between municipalities are calculated using the Haversine formula, which computes the great-circle distance between two points on a sphere given their latitude and longitude coordinates. See Table 3 for details.

Table A8: Event-Study Effects By Occupational Mobility

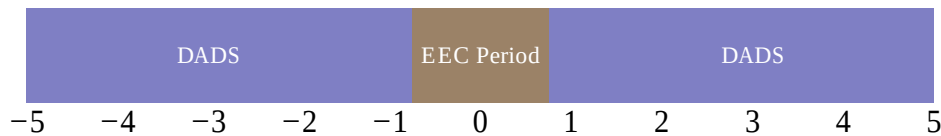
Group (EEC Period):	Part-Time			Full-Time		
Dependent Variable (in logs):	Hours (1)	Wages (2)	Earnings (3)	Hours (4)	Wages (5)	Earnings (6)
Panel A: Within-Occupation						
Mover	0.011 (0.010)	-0.025*** (0.005)	-0.013 (0.010)	0.004 (0.002)	-0.029*** (0.002)	-0.026*** (0.002)
Mover $\times$ Constrained	0.054** (0.017)	-0.013 (0.008)	0.041* (0.017)	-0.000 (0.005)	0.002 (0.004)	0.002 (0.005)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Observations	55,075	55,075	55,075	444,611	444,611	444,611
Adj. R ²	0.80	0.90	0.89	0.61	0.93	0.91
Panel B: Between-Occupation						
Mover	0.045** (0.016)	-0.063*** (0.008)	-0.018 (0.016)	-0.010** (0.003)	-0.071*** (0.002)	-0.081*** (0.004)
Mover $\times$ Constrained	0.076*** (0.026)	0.002 (0.012)	0.078** (0.026)	0.019** (0.007)	0.011* (0.005)	0.030*** (0.008)
<i>Fixed effects</i>						
Worker FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Observations	52,580	52,580	52,580	427,943	427,943	427,943
Adj. R ²	0.67	0.77	0.74	0.53	0.85	0.80

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05.

Note: This table presents estimates from event-study regressions specified in equation 10 considering within-occupation and between-occupation moves separately. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. Panel A restricts the sample to non-movers and within-occupation movers (workers who change employers while remaining in the same occupation). Panel B restricts the sample to non-movers and between-occupation movers (workers who change both employer and occupation). Occupations correspond to the 2-digit level in the PCS classification (28 values). See Table 3 for details.

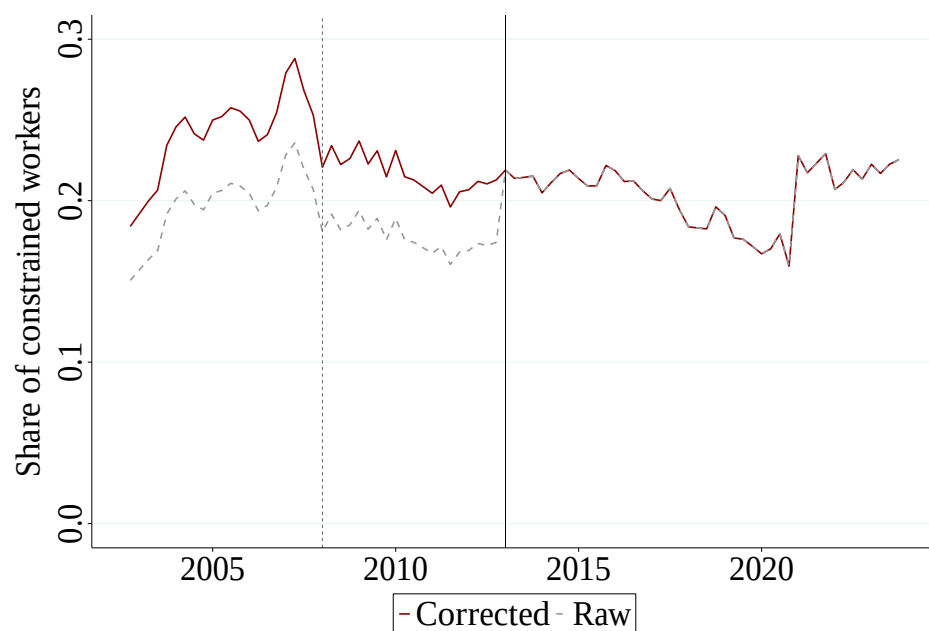
A.2 Figures

Figure A1: EEC-DADS Panel Data



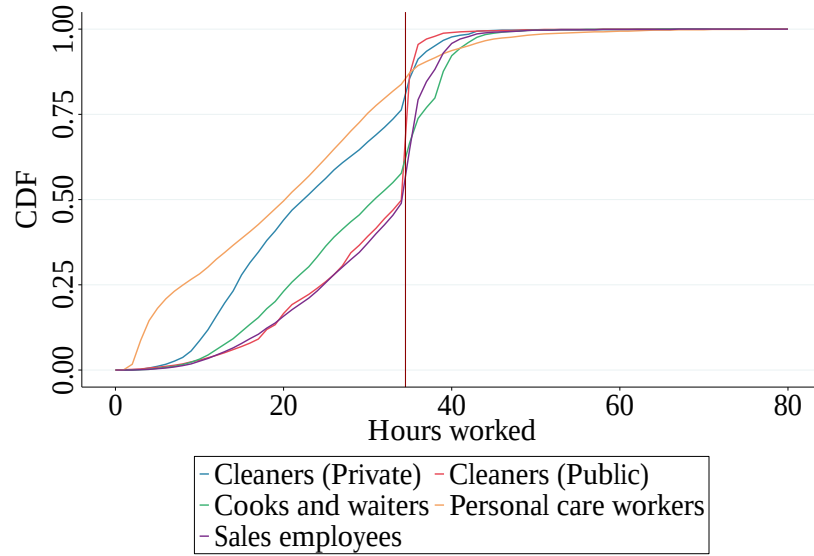
Note: This figure illustrates the structure of the newly built EEC-DADS panel data. This dataset is constructed based on the EEC-DADS dataset and the DADS panel from Godechot et al. (2025). It combines for each individual EEC-based information over 6 quarters of observation (the EEC period) and DADS-based information over the period of appearance in the panel. See Section 3.2.3 for more details.

Figure A2: Share of Workers Constrained from Above, 2003-2023



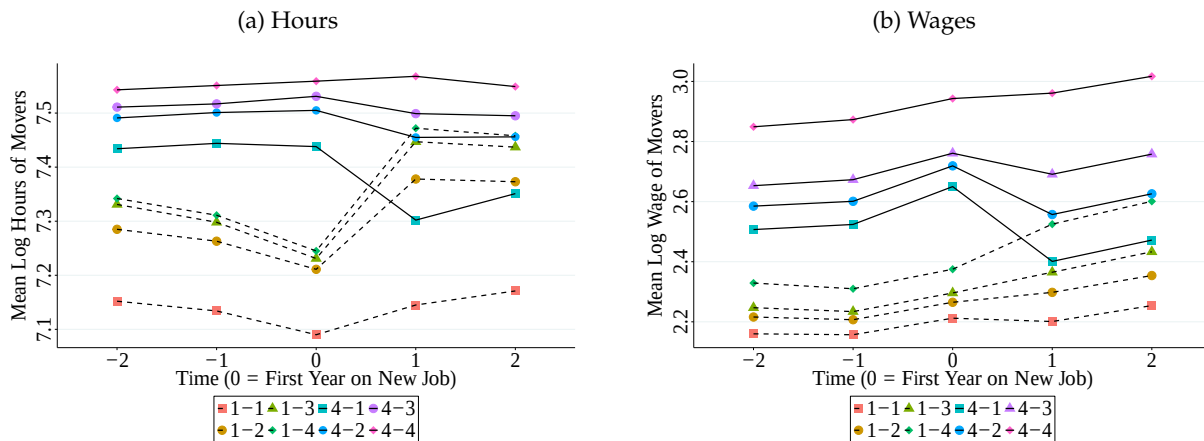
Note: The figure plots the quarterly share of employed workers reporting that they want to work more hours. The dashed grey series shows the raw data and the solid red series applies a multiplicative correction to account for a methodological break in the French Labour Force Survey (EEC) occurring in January 2013. The correction anchors on the post-2013 methodology and rescales the 2003-2012 segment so that December 2012 and January 2013 are level-consistent. The vertical dashed line corresponds to the break date.

Figure A3: Distribution of Hours Worked in IPT Occupations, 2009-2019



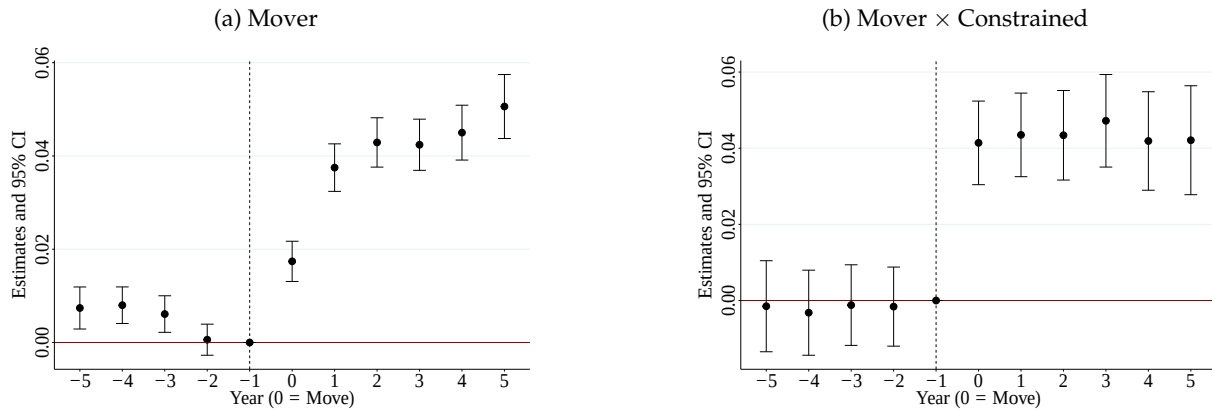
Note: The figure shows the distribution of hours worked in five occupations which concentrate nearly half of involuntary part-time workers. The vertical red line represents the 35-hour frontier between part-time and full-time work. Hours are annual paid hours from the DADS, recomputed as full-year equivalent, then as weekly-equivalent by dividing by 52. Occupations correspond to the 3-digit level (110 values) in the PCS classification. The data is restricted to years between 2009 and 2019 because the PCS classification changes in 2008 and 2020, which makes occupations not perfectly consistent in the entire window.

Figure A4: Mean Hours of Job Movers By Quartile



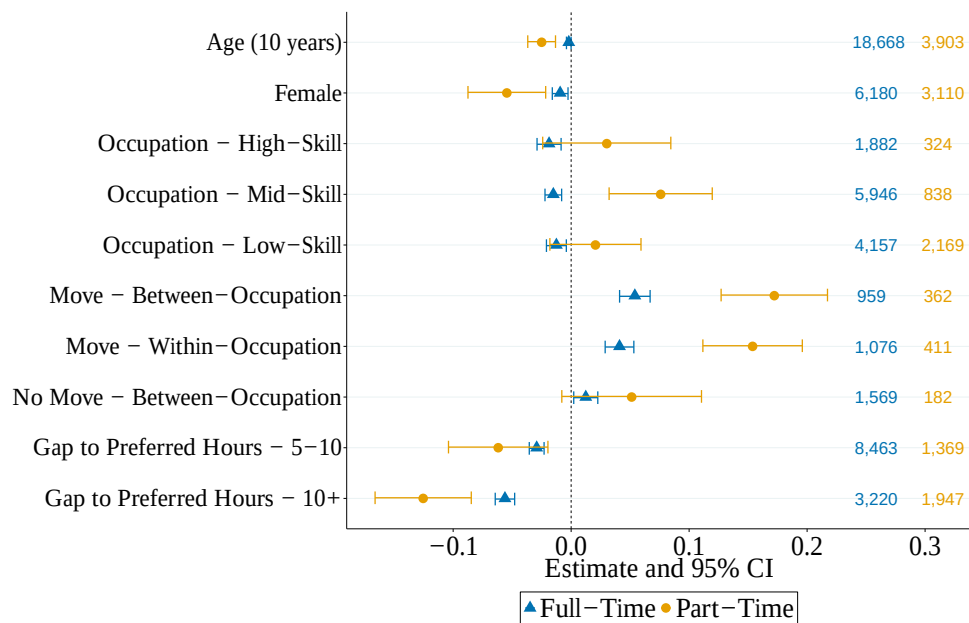
Note: Figures show mean hours (a) and hourly wages (b) of workers observed in 2005-2019 who change jobs and held the preceding job for 2 or more years, and the new job for 2 or more years, as in Card et al. (2013). The data used is the quasi-exhaustive DADS Panel derived from the procedure described in Godechot et al. (2025). Here, the job refers to the establishment with most earnings during a given year. Each job is classified into quartiles based on mean hours or wages of co-workers (quartiles are based on all full time workers in the same year).

Figure A5: Event-Study of Employer-to-Employer Moves on Hours Worked



Note: Panel (a) shows event-study coefficients for the effect of employer-to-employer moves on weekly hours worked, using never-treated individuals (non-movers) as control group. Panel (b) displays the interaction coefficients testing for differential move effects between constrained and unconstrained workers. The data used is the EEC-DADS Panel, restricted to a balanced panel between 3 years before and 3 years after the EEC period. Year 0 corresponds to the year of the employer-to-employer move, that must occur between 1 and 3 years after the EEC period. Point estimates are shown with 95% confidence intervals. Standard errors are clustered at the individual level.

Figure A6: Probability to Reach Desired Hours



Note: This figure shows coefficient estimates from separate regressions for part-time and full-time constrained workers. The dependent variable is an indicator equal to one if the worker has reached their self-reported desired hours within three years after the EEC period. All covariates are included together. Numbers on the right indicate the sample size for each group. Reference categories for categorical variables are low-skill manual workers, workers who do not move and remain in the same occupation and gaps to desired hours of less than 5 hours. Occupations correspond to the 2-digit level in the PCS classification (28 values). Error bars correspond to 95% confidence intervals.

Appendix B Measurement Issues and Data Linkage

B.1 Survey Measurement of Hours Constraints

This section provides additional background validation checks to the Labour Force Survey variables used to measure hours constraints. The paper relies on a certain interpretation of workers' answers to the corresponding questions. I argue that positive answers to the STPLC question reveal a concrete and thoughtful preference for working additional hours.

Methods to Increase Hours. Workers who report constraints from above and are available to work more hours are asked in a follow-up question (CSTPLC) about their preferred method to increase their hours with a list of possible items. 75% of these workers wish to work longer in their current job, while only 5% would rather take an additional job, 6% would opt to change jobs entirely, and 14% would do so by any means. The fact that a large majority of workers consider increasing their hours in their current working context supports the interpretation of the measure as embedded in real life context. Workers who are constrained from above thus do not appear to answer the question from a utopian perspective.

Motivations. Additional information from the EEC allows to evaluate the consistency of workers' reports with their motivations. Figure B1a shows the answers to the question *"What is the main reason you work part-time?"* (RAISTP), asked to part-time workers. Almost every involuntary part-time worker, who represent 75% of constrained (from above) part-time workers, reports being primarily limited by the absence of full-time jobs. On the other hand, part-time workers who do not wish to work longer hours typically cite personal factors like family reasons or free time as their main motivation for part-time work. The gap in that dimension between both types of workers is consistent with the measurement of constraints. Surprisingly, a non-negligible share of unconstrained workers, 20%, also report the unavailability of full-time contracts as their motivation for part-time work. This suggests a possible underestimation of the share of involuntary part-time workers. I adopt a similar approach with a question regarding motivations for another job (CREACCP). Focusing on the period between 2013 and 2023 (because of a break in series in 2013), I analyze answers to the question *"Why do you want another job?"* asked to workers who report that they want another job (either in replacement or in addition to the current one), with multiple possible answers in a list of items. Figure B1b shows that reports of a higher ideal number

of hours are associated with prospects of increased earnings and hours. Beyond the consistency of both variables, this illustrates that the desire to work long hours is likely strongly motivated by the perspective of higher earnings. Positive answers to the STPLC question thus convey ambition for a higher income at the cost of additional hours. This interpretation makes the measurement of hours constraints through the STPLC question coherent with traditional labor supply models and sets the stage for a welfare analysis of hours constraints.

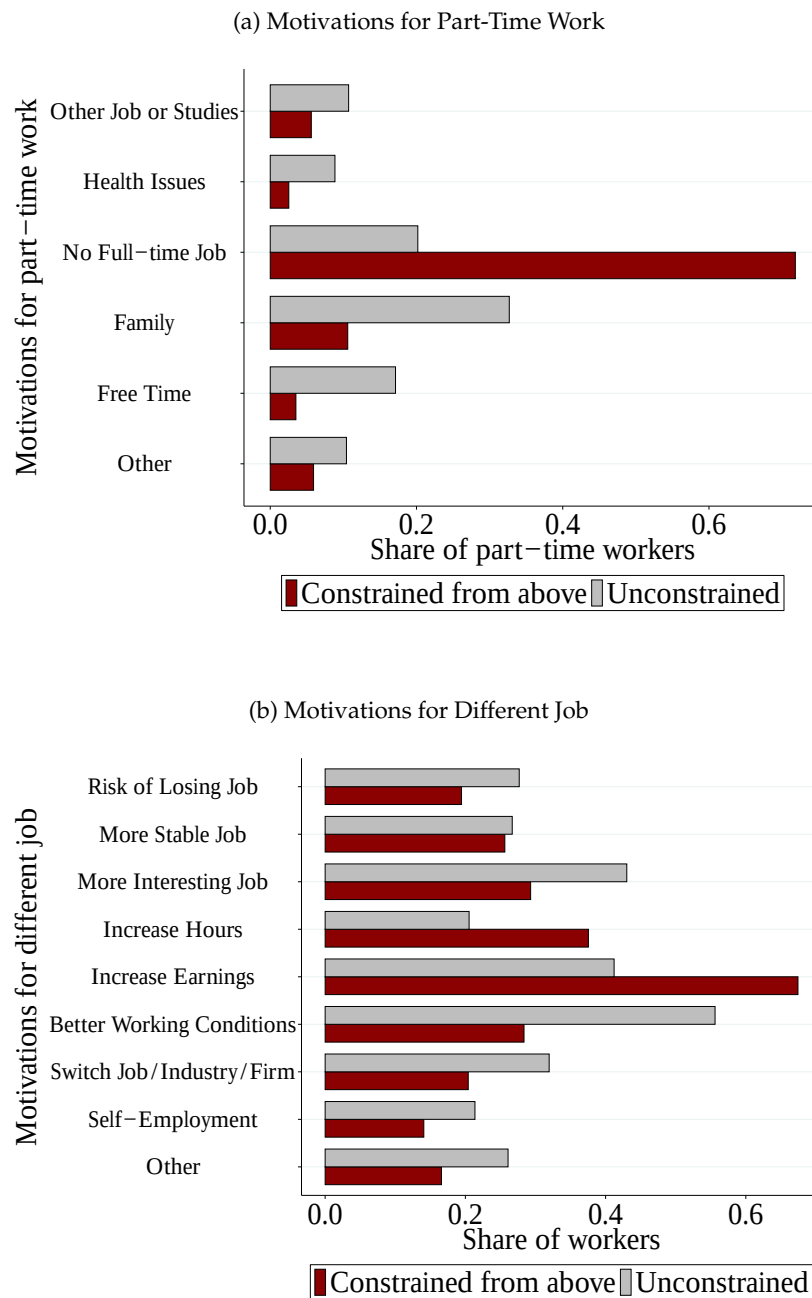
Persistence of Hours Preferences. An important question lies in understanding the stability of hours preferences over time. Individuals surveyed in the EEC are asked to report their preferences during 6 consecutive quarters at most which allows me to measure the persistence of STPLC over this period. I find that 63% of individuals who report constraints from above during their last interrogation, a definition that I use throughout Section 5, also report the same type of constraints in their first interrogation. This proportion marginally increases to 70% when only considering part-time workers but remains quite low. Additional checks suggest that the instability of preferences is not driven by changes in the employment context (employer, occupation, contract, hours worked) of the individual. This aspect of the variable is a potential threat to the results in Sections 5 and 6 which extrapolate preferences on hours over the years following the EEC period. To address this issue, I change my definition of constrained individuals to workers who report constraints both at their first and last interrogation in the EEC period. The main conclusions of the paper remain valid.

Worker Mobility. Figure B2 exploits the quarterly longitudinal structure of the EEC to examine the connection between reported preferences for hours and employer-to-employer mobility. Workers who report an ideal number of hours superior to their current one on their first interrogation switch employers significantly more than both unconstrained workers and workers who would ideally reduce their hours³⁹. Half of individuals who move report being unconstrained in their last interrogation. This finding suggests that positive answers to the STPLC question are associated with workplace disutility, hence with mobility towards alternative employers. As a result, it reinforces the interpretation of questions regarding ideal working hours as indicative of hours constraints faced by workers.

Transitions Between Constrained and Unconstrained Status. Table B1 also exploits the quarterly longitudinal structure of the EEC to examine the relationship between changes in stated hours preferences and actual labor market outcomes. Panels A and B respectively focus on workers constrained from above or unconstrained in their first interrogation. The first column of the

³⁹The result is robust to the introduction of controls on age, gender and occupation.

Figure B1: Motivations of Constrained and Unconstrained Workers

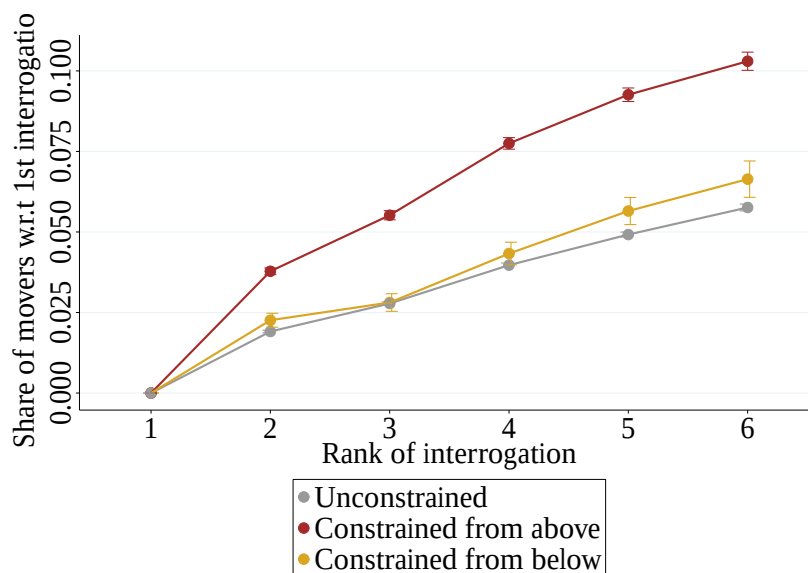


Note: These figures represent the motivations of workers to work as part-time or to look for a different job based on their constraint status in the labor market. Workers are classified as *constrained from above* or *unconstrained* based on their answer to the STPLC question in the Labour Force Survey. See 3.2.1 for more details.

(a) The figure reports answers to the question "What is the main reason you work part-time?" asked to part-time workers in the Labour Force Survey. The sample covers 144,904 workers between 2003 and 2023.

(b) The figure shows answers to the question "Why do you want another job?" asked to workers who report that they want another job (either in replacement or in addition to the current one) in the Labour Force Survey. Multiple answers are possible. The sample covers 94,573 workers between 2013 and 2023. The period is restricted to 2013-2023 for this question only in order to ensure the consistency of the variable across the years.

Figure B2: Worker Mobility in the EEC



Note: This figure represents worker mobility patterns across quarterly rounds of interrogation in the Labour Force Survey. Workers are classified as unconstrained, top-constrained, or bottom-constrained based on their answer to the STPLC question during their first interrogation. See 3.2.1 for more details. The x-axis shows the rank of interrogation while the y-axis displays the share of workers who are not in the same workplace (*SIRET*) as in their first interrogation. The sample covers 717,544 workers between 2003 and 2023.

table corresponds to the status reported in the last interrogation. Panel A shows that workers who switch from constrained to unconstrained over the EEC period experience a substantial increase in both hours and earnings compared to those who remain constrained throughout. Notably, workers who shift from being constrained from above to wanting fewer hours display even larger increases in usual hours. Panel B presents the mirror pattern: workers initially unconstrained who become constrained from above experience significant reductions in hours and earnings, while those maintaining their unconstrained status do not. Taken together, the patterns provide additional evidence for the validity of the hours preferences measure, as reported constraints correlate closely with subsequent changes in workers' labor market trajectories.

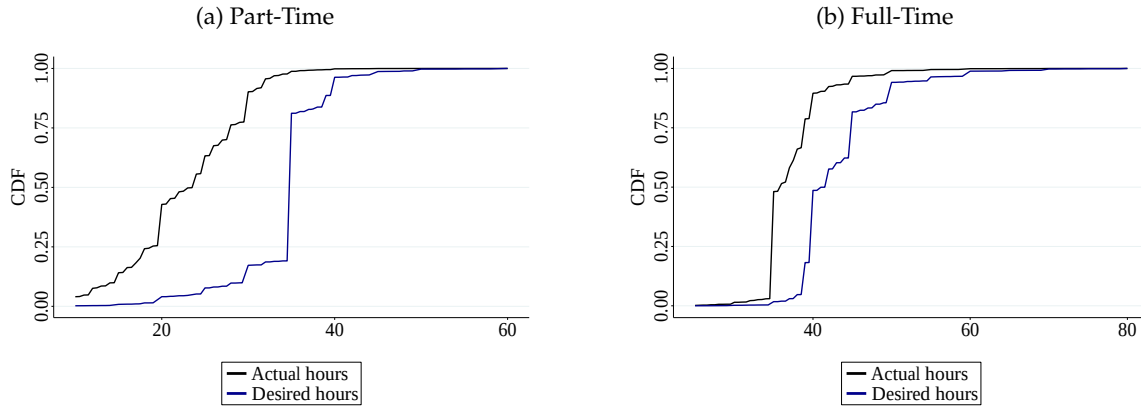
Distribution of Desired Hours. Figures B3a and B3b show the distributions of desired hours, as compared to actual hours, respectively for part-time and full-time constrained workers. Figure B3a shows a clear jump in desired hours around 35 hours for part-time workers, indicating that many constrained part-time workers want to move to full-time status. This suggests significant involuntary part-time employment among workers who cannot achieve their desired hours. Reassuringly, both panels show substantial variation in desired hours rather than excessive clustering at round numbers, e.g. like 40. This suggests that workers are genuinely reporting their true preferences rather than simply picking salient thresholds and lends credibility to the desired hours measure as capturing meaningful heterogeneity in workers' preferences.

Table B1: Changes in Hours Preferences

Panel A: Constrained from above in 1st interrogation										
Period	Usual hours			Reference hours			Earnings			N
	First	Last	Δ	First	Last	Δ	First	Last	Δ	
Constrained from above	30.1	30.3	(+0.7%)	28.0	28.0	(+0.0%)	1,288	1,305	(+1.3%)	61,322
Constrained from below	34.5	37.3	(+8.1%)	31.5	31.7	(+0.6%)	1,565	1,702	(+8.8%)	918
Unconstrained	33.3	34.8	(+4.5%)	30.6	31.7	(+3.6%)	1,496	1,572	(+5.1%)	66,479
Panel B: Unconstrained in 1st interrogation										
Period	Usual hours			Reference hours			Earnings			N
	First	Last	Δ	First	Last	Δ	First	Last	Δ	
Constrained from above	34.6	33.0	(-4.6%)	32.1	30.4	(-5.3%)	1,523	1,491	(-2.1%)	38,050
Constrained from below	41.1	41.5	(+1.0%)	37.5	36.4	(-2.9%)	2,237	2,277	(+1.8%)	8,010
Unconstrained	38.4	38.4	(+0.0%)	35.4	34.8	(-1.6%)	1,932	1,934	(+0.1%)	434,168

Note: This table presents changes in hours and earnings across changes in hours preferences based on Labour Force Surveys pooled between 2003 and 2023. Labels in the first column correspond to the worker's status in their last interrogation. Panel A restricts the sample to workers who report wanting to work more in their first interrogation. Panel B restricts the sample to workers who report not wanting to work more nor less in their first interrogation. Reference hours correspond to hours worked during the reference week of the Labour Force Survey. Earnings correspond to net monthly labor earnings.

Figure B3: Distribution of Actual and Desired Hours, EEC 2003-2023



Note: The figures show the distributions of actual and desired hours worked for constrained workers based on Labour Force Surveys pooled between 2003 and 2023. The sample has been restricted to constrained employees with hours worked above 10 and an hourly wage between 0.8 and 1000 times the hourly minimum wage.

B.2 Measurement of Hours Worked in French Administrative Data

This section aims to assess the reliability of working hours reported in the French matched employer-employee data (DADS). Those administrative datasets cover almost the entire salaried workforce and have been widely used in labor market studies. Reporting hours in the DADS is mandatory and the information is used to manage and administer unemployment insurance and social benefits. To check the validity of hours, I rely on the methodology of [Lachowska et al. \(2022\)](#), applied to the administrative data of the State of Washington, and extend it by incorporating elements

specific to the French context. This methodology has become a standard procedure to evaluate the quality of hours data (see e.g. [Labanca and Pozzoli \(2022\)](#)).

As in the original paper, I first compare the distributions of hours worked in the administrative data and the Labour Force Survey, including on a common sample of workers. Although the definitions of hours worked differ according to the sources, this comparison highlights the degree of reliability of the variable according to the type of situation. Second, I measure the correlation between the evolution of annual earnings and hours worked for individuals who remain in the same job over two successive years of data, denoted as the “signal-to-noise” ratio in [Lachowska et al. \(2022\)](#). Assuming that their hourly wage should remain constant over time, the estimated coefficient should be relatively close to 1 (0.8 in the original paper). Third, I examine to what extent the lower end of the hourly wage distribution follows the evolution of the hourly minimum wage (the *Smic* in France) over time.

B.2.1 Hours Distributions in Admin and Survey Data

The first part compares the working hours distributions reported in the DADS and the Labour Force Survey, which is the main source of survey data for working time studies. I consider data for the years 2005, 2010, 2015, and 2019, covering the entire study period (2003-2023) while maintaining a manageable dataset size. Some restrictions are applied to the DADS data to limit measurement errors: workers aged between 18 and 64, in ordinary jobs of more than 120 hours and 60 days. I also keep one observation per person-year, the one with highest annual earnings to maintain consistency with the EEC definition of working hours *in the primary job*. Despite these adjustments, the comparison exercise has notable limitations.

First, the definition of hours worked is not the same across sources. The DADS data records the total number of paid hours for a given employer over the year. This definition is closer to contractual working time including standard hours, overtime, and paid absences. The EEC data, on the other hand, asks individuals about their *usual* hours worked (including overtime) per week. In short, the EEC aims to estimate the actual working time of the individual, including times that do not necessarily result in remuneration, while the DADS reflects the employer’s perspective on the official hours of their employees. Annual hours from the DADS are recomputed as weekly hours using the starting and ending dates of employment. Second, the scope of the two sources differs. The DADS provides information on all salaried workers, while the EEC draws its sample from all individuals living in ordinary housing, including inactive or non-salaried individuals. I retain only salaried workers in this sample, although they are not representative of the entire population. To address potential sampling issues, I also provide distributions of both types of hours from the common EEC-DADS sample introduced in Section [3.2.3](#).

Figures B4a and B4b show the distributions of hours respectively from the DADS and the EEC. It is clear that the 35-hour peak, the standard duration in France, is more pronounced in administrative data, compensated by an over-representation of declarations between 37 and 45 hours in the EEC. Figure B5 compares both distributions in the common EEC-DADS sample. The profiles are very close to the original figures which suggests that the gap in hours is not driven by representativeness issues. Figure B6 represents average DADS paid hours at each level of EEC usual hours. Values are very close to the $x = y$ line in the part-time space (below 35) while in the full-time case both measures show very limited correlation.

Two particularities of the French working time regulation system can help to explain the difference. First, *forfait jours* contracts are a French flexible work arrangement which entails that working time is measured in terms of annual working days rather than in weekly hours. For workers subject to this arrangement, about 15% of the workforce in 2019, the employer is not required to monitor hours worked, nor to report them in their DADS record. The absence of hours worked is circumvented by Insee through recoding to an arbitrary number of hours. Since 2017, a variable of the DADS dataset (UNITMESUREREF) allows to identify workers with a *forfait jours* contract. They turn out to be mainly attributed either 1820 or 2200 annual hours depending on the year of data. *Forfait jours* contracts are excluded from the sample in the majority of the paper and their removal is made explicit. Second, the gap in hours distributions between DADS and EEC could also be due to the *Réduction du Temps de Travail* (RTT) scheme. This arrangement grants rest days or half-days to an employee if the working time exceeds 35 hours per week (up to 39 hours weekly). Then, an employee who has a contract with 39 hours per week would probably report 39 usual hours in the EEC but their total number of hours over the year would correspond to a 35-hour equivalent. Apart from those two mechanisms, the gap in hours may reflect that workers on the right-side of the graph tend to over-report their hours.

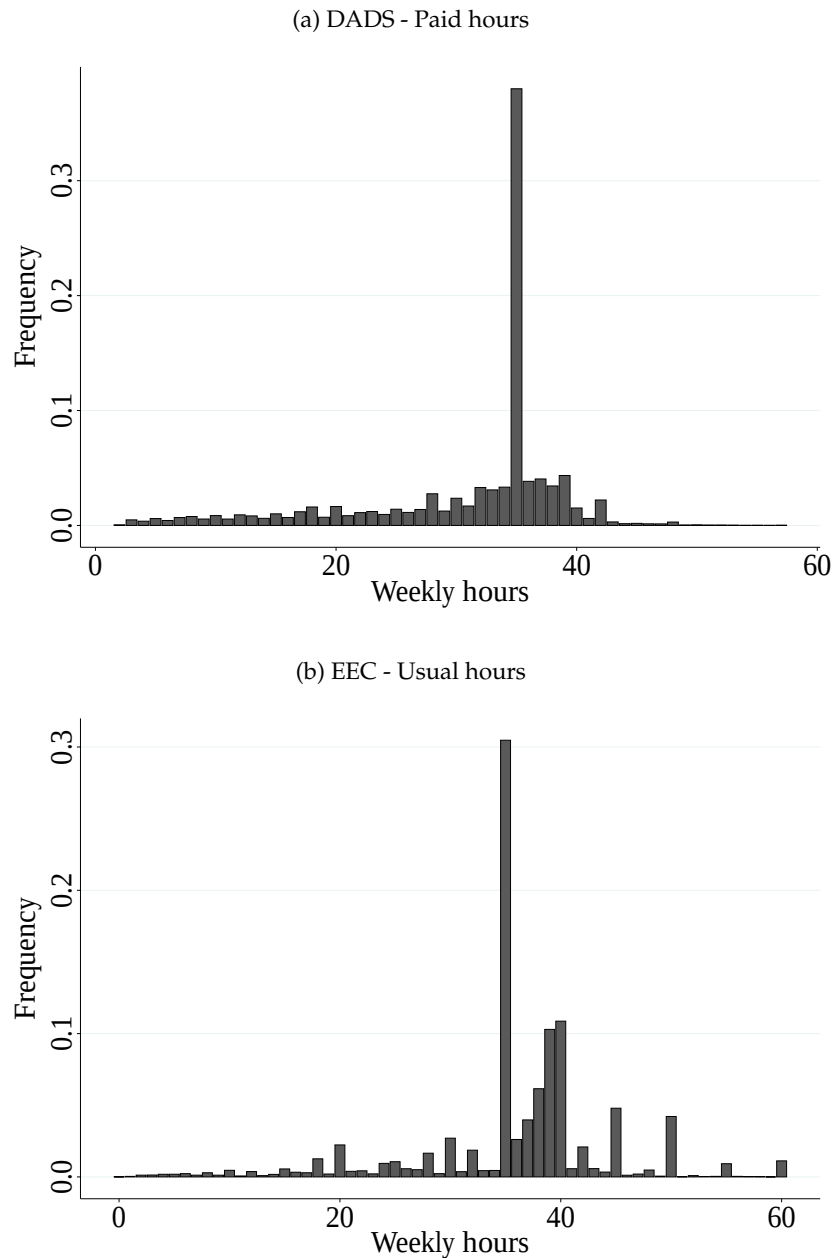
B.2.2 Signal-to-Noise Regressions

The hourly wage of a worker paid through an hours contract can be considered to fluctuate stochastically around a fixed hourly rate. This observation from Lachowska et al. (2022) suggests that a simple regression of the evolution of annual labor earnings in logs between two periods $[\Delta \ln(earn_{it})]$ on the equivalent change in terms of hours worked $[\Delta \ln(hrs_{it})]$ provides a test of the reliability of the measurement of hours:

$$\Delta \ln(earn_{it}) = \alpha + \beta \Delta \ln(hrs_{it}) + \varepsilon_{it}$$

The test is not as relevant in the DADS yearly data as it is in the original Washington state quarterly data because hourly wages are more likely to increase e.g. due to promotion. Still, if hours

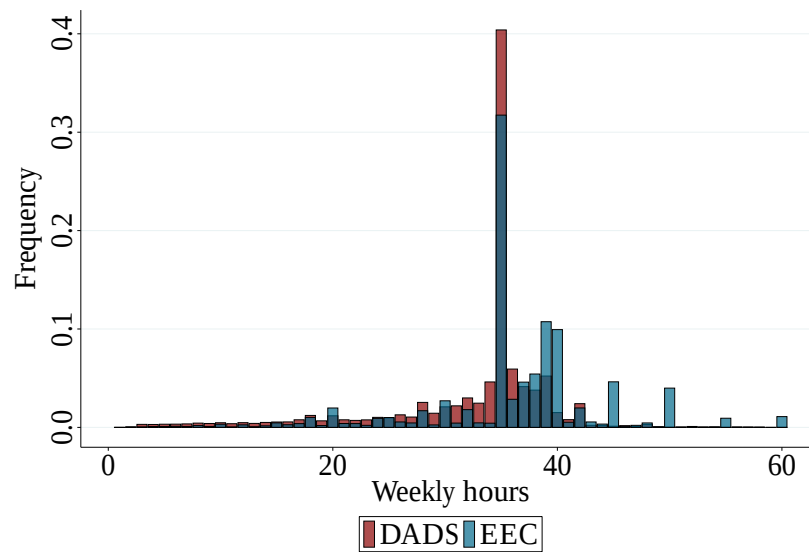
Figure B4: Distributions of Weekly Hours, EEC vs DADS, 2005, 2010, 2015 and 2019



Note: These figures show the distributions of weekly working hours in the DADS and the French Labour Force Survey (EEC) in years 2005, 2010, 2015 and 2019. The sample includes salaried workers aged 15-64 in ordinary jobs of more than 120 hours and 60 days. Annual paid hours from the DADS are recomputed as full-year equivalent using the employment duration and then divided by 52 and rounded to obtain weekly hours. Values above 60 are not displayed.

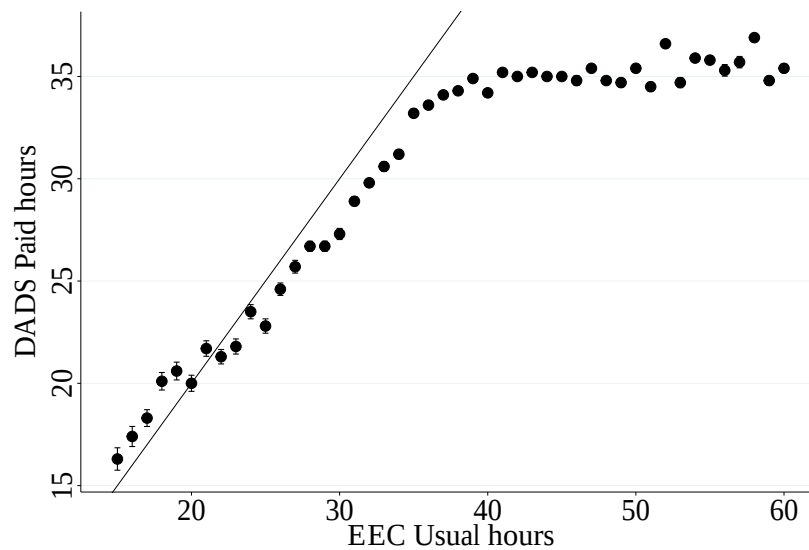
are measured accurately, the estimates of the coefficient on the evolution of hours (β) should be arbitrarily close to 1 for hourly paid workers. On the other hand, if hours are measured with significant error, the coefficient from this simple regression is expected to be *attenuated*. Using the DADS yearly files between 2018 and 2019, I retain workers who stay in the same company and the

Figure B5: Distributions of Weekly Hours, EEC-DADS



Note: This figure shows the distributions of weekly working hours in the EEC-DADS dataset between 2003 and 2023. The sample includes salaried workers aged 15-64 in ordinary jobs of more than 120 hours and 60 days. Annual paid hours from the DADS are recomputed as full-year equivalent using the employment duration and then divided by 52 and rounded to obtain weekly hours. Values above 60 are not displayed.

Figure B6: Comparison of Hours Worked, EEC-DADS



Note: This figure shows the relation between DADS paid hours and EEC usual hours in the EEC-DADS dataset between 2003 and 2023. Values correspond to the average of DADS paid hours at each (rounded) level of EEC usual hours. The diagonal line has slope 1. The sample includes salaried workers aged 15-64 in ordinary jobs of more than 120 hours and 60 days. Annual paid hours from the DADS are recomputed as full-year equivalent using the employment duration and then divided by 52 and rounded to obtain weekly hours. EEC weekly hours below 15 and above 60 are not displayed. Error bars correspond to the coefficient of variation.

same occupation across years and estimate the coefficient β in different specifications.

Table B2: Signal-to-Noise Coefficients between Earnings and Hours, 2018-2019

Dependent variable:	$\Delta \ln(earn_{it})$			
	(1)	(2)	(3)	(4)
$\Delta \ln(hrs_{it})$	0.684 (0.000)	0.684 (0.001)	0.692 (0.002)	0.692 (0.001)
Employer FE	No	No	Yes	Yes
Std. Errors	Standard	Clustered at worker level	Standard	Clustered at worker level
Observations	12,608,316	12,608,316	12,608,316	12,608,316
Adj. R^2	0.49	0.49	0.62	0.62

Note: This table shows the results of the signal-to-noise regression using different specifications. The sample is composed of workers from the DADS 2018 annual dataset who stay with the same primary employer and the same occupation in 2019. Standard errors are in parentheses.

The results show a coefficient slightly below 0.7 stable across specifications. Complementary results by industry show strong homogeneity in signal-to-noise coefficients. [Lachowska et al. \(2026\)](#) consider that a coefficient equal to 0.8, in an area with 63% hourly-paid workers indicates low measurement error in hours. The lower proportion of hourly-paid workers in France coupled with the yearly nature of the data partly explain the lower coefficient and suggest that the DADS data could have a measurement quality similar to Washington state data.

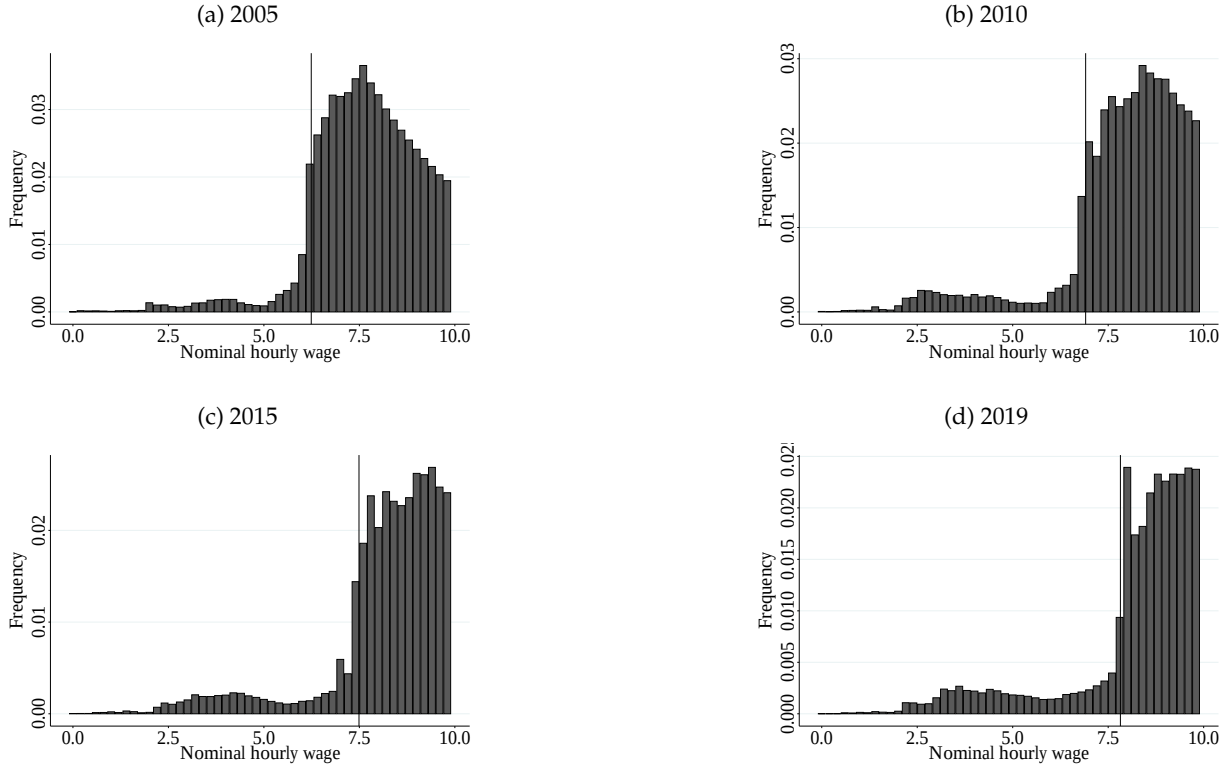
B.2.3 Minimum Wage Changes and Nominal Wage Distributions

A simple test of the reliability of working hours, at least for low-wage employees, is to observe whether the distribution of nominal hourly wages reflects the variations in the net hourly minimum wage (the *Smic horaire*) over time. The hourly minimum wage is indexed to the inflation rate measured for workers below the first quintile of the earnings distribution. Figure [B7](#) presents the results for 2005, 2010, 2015, and 2019. The distribution of hourly wages exhibits a shift in the concentration of wages over time that corresponds to variations in the net hourly minimum wage. This pattern would not be observable in a database with significant measurement error. In particular, this exercise rules out systematic, time-invariant reporting.

B.3 Matching EEC and DADS

This section describes the procedure for linking the French Labour Force Survey (EEC) with the French matched employer-employee dataset (DADS) at the individual level. The general approach consists of finding the individuals surveyed by the EEC in the DADS exhaustive records. The sample only includes perfectly identified individuals i.e. who appear in a 1-to-1 match. This sample is then connected to the DADS panel obtained by following the procedure of [Godechot](#)

Figure B7: Distribution of Hourly Wages and Minimum Wage



Note: These figures represent the distributions of nominal hourly wages around the net hourly minimum wage (the *Smic horaire*) in 2005, 2010, 2015 and 2019. Nominal hourly wages are obtained by dividing net annual earnings by the annual number of hours worked in the DADS. Black vertical lines correspond to the net hourly minimum wage for the respective year. The values for the hourly minimum wage (in current euros) are 6.24 (a), 6.91 (b), 7.49 (c) and 7.82 (d). Distributions are divided into intervals of 0.20 euros and values above 10 euros are not displayed.

[et al. \(2025\)](#). This ultimately leads to an innovative EEC-DADS Panel combining 6 quarters of observation in the EEC with the worker's employment history.

The challenge of matching the French Labour Force Survey with the DADS lies in the absence of a common individual identifier. However, both datasets share a number of variables that are supposed to be consistent: age, sex, occupation, part-time/full-time work, birth department, birth month (only between 2009 and 2012), residence municipality, and establishment's ID (*SIRET*). Table B3 reports the names of the variables. The *SIRET* is the key variable that makes this match feasible, as it considerably reduces the number of individuals with the same characteristics and therefore limits the risk of "1-to-many" matches. Age (at the end of the year) and sex are never missing and have no reason to differ across sources for a given individual. The combination of these three variables is the base of the matching procedure and already provides a strong identifier, as "1-to-many" matches would only occur if there are at least two individuals of the same age and gender who work in the same workplace during the same year. Other variables are more

subject to misreporting and are thus at some point recoded or removed from the list of matching variables. The idea is to use all common variables to build a correspondence table between individual identifiers of both sources for each year. Importantly, the EEC information concerns the primary job of the employee, which means that the linkage is realized through this employment spell. But, once an individual has been matched and appears in the correspondence table, all their employment spells during the year can be recovered from the DADS.

Table B3: Variables for EEC-DADS Matching

Variable	EEC	DADS	Description
Establishment's ID	SIRET	SIRET	The SIRET is a unique identifier for establishments in France. In the EEC, the variable is coded by Insee based on the establishment's address reported by the individual. In the DADS, it is directly reported by HR services.
Age	AG	AGE	Age of the individual at the end of the year.
Sex	SEXE	SEXE	Gender of the individual (male/female classification).
Occupation	CSTOT, P	CS, PCS	The French occupational classification system changed in 2008. 2-digit variables are used prior to 2009 (CSTOT in the EEC and CS in the DADS), while the 4-digit current system variables (P in the EEC and PCS in the DADS) are used afterwards.
Working Time Status	TPPRED	CPFD	Employment status (full-time or part-time).
Birth County	DNAI	DEP.NAISS	French <i>département</i> (101 values) where the individual was born.
Birth Month (2009-2012)	NAIM	MOIS.NAISSANCE	Month of birth, available only for the specified period in the DADS.
Municipality of Residence	DEPCOM	COMR	French municipality, identified by the <i>Code commune</i> , where the individual resides.

Note: This table contains the names and description of variables used to link the EEC and DADS datasets. Variable names correspond to the 2019 version of each dataset (except for occupation and birth month). Multiple revisions between 2003 and 2023 have induced changes in the names and content of several variables. These changes are taken into account in the cleaning process in order to build a version consistent over the years.

The procedure starts with some minor cleaning of the EEC to make some variables consistent with their DADS equivalent. All quarterly files between 2003 and 2023 are pooled, restricting the sample to employees whose *SIRET* information is not missing - about 81% of the sample (see Table A1). Likewise, the regional DADS files are pooled to build yearly DADS datasets with the minimum

required number of variables (the individual ID *IDENT_S* and the matching variables). The match is then performed in multiple steps with different subsets of the set of matching variables in order to optimize "1-to-1" matches. Here are the 11 sets of variables used through the procedure (using the DADS labels reported in Table B3):

- **Set 1:** SIRET, AGE, SEXE, PCS, CPFD, DEP_NAISS, COMR
- **Set 2:** SIRET, AGE, SEXE, PCS, CPFD, COMR
- **Set 3:** SIRET, AGE, SEXE, PCS, CPFD, DEP_NAISS
- **Set 4:** SIRET, AGE, SEXE, CPFD, DEP_NAISS, COMR
- **Set 5:** SIRET, AGE, SEXE, CS, CPFD, DEP_NAISS, COMR
- **Set 6:** SIRET, AGE, SEXE, CS, CPFD, COMR
- **Set 7:** SIRET, AGE, SEXE, CS, CPFD, DEP_NAISS
- **Set 8:** SIRET, AGE, SEXE, CS 1st digit, DEP_NAISS, COMR
- **Set 9:** SIRET, AGE, SEXE, CPFD, DEP_NAISS
- **Set 10:** SIRET, AGE, SEXE, DEP_NAISS
- **Set 11 (2009-2012):** SIRET, AGE, SEXE, MOIS_NAISSANCE

The match is run with a R script called [2_eecdads_id.R](#) at the yearly level using clean and appended EEC and DADS files. At each step, I remove from both datasets individuals who have already been matched in order to avoid double matching. The complete process yields a correspondence table for each year between unique identifiers from the EEC (*IDENT* $\times$ *NOI*) and the DADS (*IDENT_S*). Using the correspondence table, one can retrieve all observations of identified individuals in the EEC and the DADS yearly files to build the EEC-DADS sample.

Appendix C Theory

C.1 Lewis-Rosen Model Proofs

Static Bargaining over Hours and Wages (Equations 2 and 3). As in Appendix B of [Lachowska et al. \(2026\)](#), I provide the derivations for the expression of bargained hours and wages. Under imperfect competition, bargained hours and wages solve the following maximization problem:

$$(w_{ij}^b, h_{ij}^b) = \operatorname{argmax}_{w, h} u_i(e, h) \text{ s.t. } \Pi_j(h, \theta_i, w) = k_j$$

The Lagrangian associated with this problem is:

$$\mathcal{L}(w, h) = u_i(wh, h) + \lambda [R_j(h, \theta_i) - wh - k_j]$$

The first-order conditions of this problem are:

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \Rightarrow \frac{\partial u_i}{\partial e} \cdot h - \lambda h = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial h} = 0 \Rightarrow \frac{\partial u_i}{\partial e} \cdot w + \frac{\partial u_i}{\partial h} + \lambda \left[\frac{\partial R_j}{\partial h} - w \right] = 0 \quad (2)$$

(1) yields the equality $\frac{\partial u_i}{\partial e} = \lambda$, which can be plugged into (2):

$$\frac{\partial u_i}{\partial e} \left[w + \frac{\partial R_j}{\partial h} - w \right] + \frac{\partial u_i}{\partial h} = 0 \Rightarrow -\frac{\frac{\partial u_i}{\partial h}}{\frac{\partial u_i}{\partial e}} = \frac{\partial R_j}{\partial h}$$

which corresponds to Equation 2.

Equation 3 then directly derives from the profit constraint, using h_{ij}^b , the solution for bargained hours:

$$R_j(h_{ij}^b, \theta_i) - wh_{ij}^b - k_j = 0 \Rightarrow w_{ij}^b = \frac{R_j(h_{ij}^b, \theta_i)}{h_{ij}^b} \left[1 - \frac{k_j}{R_j(h_{ij}^b, \theta_i)} \right].$$

Optimal Search Intensity (Equation 8). The worker employed at firm j chooses search intensity $s_i \geq 0$ to maximize the lifetime utility $V_{ij}(u)$, which satisfies the Bellman equation:

$$(\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(T_j)) V_{ij}(u) = u - c(s_i) + \delta V_i^u + \lambda(s_i, \theta_i) \int_{T_j}^{\bar{T}} V_{ix} \left(u_{ix}^{\text{poaching}} \right) d\Gamma(x)$$

Solving for $V_{ij}(u)$ yields:

$$V_{ij}(u) = \frac{u - c(s_i) + \delta V_i^u + \lambda(s_i, \theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{\text{poaching}}) d\Gamma(x)}{\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(T_j)}$$

Since $\lambda(s_i, \theta_i) = \lambda \cdot \phi(\theta_i) \cdot s_i$, we have $\partial\lambda/\partial s_i = \lambda \cdot \phi(\theta_i)$. Differentiating $V_{ij}(u)$ with respect to s_i and setting the result to zero, the first-order condition $\partial V_{ij}/\partial s_i = 0$ is equivalent to:

$$\begin{aligned} & \left[-c'(s_i) + \lambda\phi(\theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{\text{poaching}}) d\Gamma(x) \right] [\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(T_j)] \\ &= \left[u - c(s_i) + \delta V_i^u + \lambda(s_i, \theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{\text{poaching}}) d\Gamma(x) \right] \lambda\phi(\theta_i) \bar{\Gamma}(T_j) \end{aligned}$$

Dividing both sides by $\beta + \delta + \lambda(s_i, \theta_i) \bar{\Gamma}(T_j)$ and recognising that the right-hand side equals $V_{ij}(u) \cdot \lambda\phi(\theta_i) \bar{\Gamma}(T_j)$:

$$-c'(s_i) + \lambda\phi(\theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{\text{poaching}}) d\Gamma(x) = V_{ij}(u) \cdot \lambda\phi(\theta_i) \bar{\Gamma}(T_j)$$

Using $\bar{\Gamma}(T_j) = \int_{T_j}^{\bar{T}} d\Gamma(x)$, the right-hand side becomes $\lambda\phi(\theta_i) \int_{T_j}^{\bar{T}} V_{ij}(u) d\Gamma(x)$, so that:

$$c'(s_i) = \lambda\phi(\theta_i) \int_{T_j}^{\bar{T}} V_{ix}(u_{ix}^{\text{poaching}}) d\Gamma(x) - \lambda\phi(\theta_i) \int_{T_j}^{\bar{T}} V_{ij}(u) d\Gamma(x)$$

Factoring $\lambda \cdot \phi(\theta_i)$ and combining the two integrals over the common domain $[T_j, \bar{T}]$ yields the optimal search intensity condition in Equation 8:

$$c'(s_i) = \lambda \cdot \phi(\theta_i) \int_{T_j}^{\bar{T}} \left[V_{ix}(u_{ix}^{\text{poaching}}) - V_{ij}(u) \right] d\Gamma(x) \quad (13)$$

C.2 Willingness-To-Pay

Labor Supply Elasticity. The individual chooses h to maximize $u_i(w, h)$. The first-order condition with respect to h is:

$$\frac{\partial u_i}{\partial h} = w - \varepsilon_i \mu h^{\mu-1} = 0 \quad (14)$$

Solving for h yields the Marshallian labor supply function:

$$h^*(w) = \left(\frac{w}{\varepsilon_i \mu} \right)^{\frac{1}{\mu-1}} \quad (15)$$

Taking the natural logarithm of (15):

$$\ln h^* = \frac{1}{\mu-1} \ln w - \frac{1}{\mu-1} \ln(\varepsilon_i \mu) \quad (16)$$

Differentiating with respect to $\ln w$ yields the Marshallian (uncompensated) wage elasticity of labor supply:

$$\eta \equiv \frac{\partial \ln h^*}{\partial \ln w} = \frac{1}{\mu-1} \quad (17)$$

The elasticity η is constant across workers. A higher value of μ implies a more rapidly increasing marginal disutility of work, and thus a less elastic labor supply response to wages.

Alternative Definitions. Here, I compare the main definition of willingness-to-pay (WTP) introduced in Section 6 with the definition from [Lachowska et al. \(2026\)](#). In particular, I show that the latter is always superior to the former in the case where an individual who is constrained from above.

For simplicity, denote $u_0 = u(h_0, w_0)$ and $u^* = u(h^*, w_0)$, the two levels of utility that drive the estimation. Starting from the functional form of worker utility, for a given utility level u and hours h , the hourly wage is:

$$w(u, h) = \frac{u + \varepsilon h^\mu}{h}$$

Using this expression, both definitions of the WTP can be rewritten as:

$$\begin{aligned} \theta &= w_0 - w(u_0, h^*) = \frac{u^* + \varepsilon h^{*\mu}}{h^*} - \frac{u_0 + \varepsilon h^{*\mu}}{h^*} = \frac{u^* - u_0}{h^*} \\ \theta_{alt} &= w(u^*, h_0) - w_0 = \frac{u^* + \varepsilon h_0^\mu}{h_0} - \frac{u_0 + \varepsilon h_0^\mu}{h_0} = \frac{u^* - u_0}{h_0} \end{aligned}$$

In the case where the individual is constrained from above, h_0 is lower than h^* , by definition. Hence, $\theta_{alt} > \theta$ is always true for a given constrained worker.